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Problems in Problem-Based Design Engineering Education

Towards a Multidisciplinary Designerly Educational Approach



LINKÖPING UNIVERSITY

Juan Felipe Ruiz M

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Juan Felipe Ruiz Muñoz

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A mi amada familia

"The first component of intelligence
involves effective adaptation to an environment."
James G. March

Abstract

Modern industry faces complex and 'wicked' problems that require engineering professionals to go beyond traditional natural science-based linear problem-solving approaches and adopt collaborative, multidisciplinary, and iterative problem-solving strategies. To tackle these kinds of problems, organizations are increasingly turning to design problem-solving methods based on the designer's way of thinking, acting, and doing. Designers have a distinctive ability to deal with poorly defined, ambiguous, or "wicked" problems by emphasizing iterative exploration of both the problem and the solution spaces. They do this through design reasoning patterns that involve constant iteration and temporary solutions. This shift towards designerly ways of problem-solving has, in turn, had an effect on engineering education, where there has been a significant shift towards educational models that utilize design methodologies to engage students in immersive problem-solving experiences. One challenge for educators who utilize models based on designerly thinking is to create structures that actually support the learning objectives, and the development of student skills that are rooted in design reasoning and acting, and not merely in design tools. Another challenge is to support collaboration across multiple areas that traditionally had clear boundaries.

This thesis studies practices utilized by educators in problem-based designerly education to understand the underlying mechanisms and theoretical underpinnings of problem exploration in multidisciplinary education. Additionally, this thesis aims to explore and discuss the same processes and methods in the context of multidisciplinary education and design objects that can support collaboration across boundaries.

This thesis's key contributions are the exploration and discussion of aspects of problem exploration, framing, and reframing in a designerly problem-based multidisciplinary educational environment, as well as the challenges and difficulties that educators and students encounter in the process of exploring problems and collaborating and crossing disciplinary boundaries with participants from multiple disciplines. To do so, this thesis first explores the importance that (the framing of the) design brief has in the problem-solving process. Furthermore, the design briefs are discussed as boundary objects that serve a crucial role in negotiation, communication,

and coordination tools between stakeholders. Second, the importance of the reflective process that follows the idea generation and prototype-building activities are discussed as an aspect of an educational model that allows participants to explore problems and avoid design fixation. Moreover, these objects are discussed based on their function as disciplinary boundary crossing objects and as an aid in negotiation, and collaboration objects in problem exploration. Thirdly, methods and processes for assessment of student characteristics and skills are discussed, where tensions and trade-offs between self-reporting and observer-based methods are studied and explored. These methods then serve as boundary objects in the discussions between teaching teams in the student team formation process. Furthermore, team building and specifically the process of trust-building and objects that aid in boundary-crossing collaboration and communication to develop trust between students are also discussed.

Sammanfattning

Den moderna industrin står inför komplexa och så kallade "lömska"(eng. wicked) problem som kräver att ingenjörer går längre än traditionella naturvetenskapsbaserade linjära problemlösningsmetoder och antar samarbetande, multidisciplinära och iterativa problemlösningsstrategier. För att ta itu med den här typen av problem vänder sig organisationer i allt högre grad till designproblemlösningsmetoder baserade på designerns sätt att tänka, agera och göra.

Designers har en utmärkande förmåga att hantera dåligt definierade, tvetydiga och "lömska" problem genom att betona iterativ utforskning av både problemet och lösningsutrymmena. De gör detta genom designresonemangsmönster som involverar konstant iteration och tillfälliga lösningar. Denna förändring mot designmässiga sätt att lösa problem har i sin tur haft en effekt på ingenjörsutbildningen, där det har skett en betydande förändring mot utbildningsmodeller som använder designmetoder för att engagera studenter i verklighetsnära problemlösningsupplevelser. En utmaning för pedagoger som använder modeller baserade på designtänkande är att skapa strukturer som faktiskt stödjer lärandemålen och utvecklingen av studenters färdigheter som är förankrade i designresonemang och agerande, och inte bara i designverktyg. En annan utmaning är att stödja samarbete över flera områden som traditionellt har haft tydliga gränser.

Denna avhandling studerar den praxis som används av lärare i problembaserad designutbildning för att förstå de underliggande mekanismerna och teoretiska grunderna för problemutforskning i multidisciplinär utbildning. Dessutom syftar denna avhandling till att utforska och diskutera samma processer och metoder inom ramen för multidisciplinär utbildning och skapa designobjekt som kan stödja samarbete över gränser.

Den här avhandlingens nyckelbidrag är utforskandet och diskussionen av aspekter av problemutforskning, inramning och omformulering i en designmässigt problembaserad multidisciplinär pedagogisk miljö, såväl som de utmaningar och svårigheter som lärare och studenter möter i processen att utforska problem och samarbeta när disciplinära gränser korsas med deltagare från flera discipliner. För att göra det undersöker denna avhandling först vilken betydelse (utformningen av) "designbriefs"

har i problemlösningsprocessen. Vidare diskuteras "designbriefs" som gränsobjekt som har en avgörande roll i förhandlings-, kommunikations- och samordningsverktyg mellan intressenter. För det andra diskuteras vikten av den reflekterande process som följer på idégenereringen och prototypbyggande aktiviteter som en aspekt av en utbildningsmodell som tillåter deltagarna att utforska problem och undvika designfixering. Dessa objekt diskuteras också utifrån sin funktion som disciplinära gränsöverskridande objekt och som hjälpmedel vid förhandling, och samverkansobjekt vid problemutforskning. För det tredje diskuteras metoder och processer för bedömning av studenters egenskaper och färdigheter, där spänningar och avvägningar mellan självrapportering och observatörsbaserade metoder studeras och utforskas. Dessa metoder fungerar sedan som gränsobjekt i diskussionerna mellan lärarlag i teamformeringsprocessen. Vidare diskuteras teambuilding och specifikt processen att bygga upp tillit och objekt som hjälper till i gränsöverskridande samarbete och kommunikation för att utveckla tillit mellan studenterna.

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Publications included in the thesis

Licentiate Thesis:
Ruiz, J (2020) The Problem of Formulating Design Problems; A Typology of Design Briefs.
Monograph Licentiate thesis.

Paper I
Ruiz, J., & Wever, R. (2024). Design team formation using self-assessment and observer-assessment techniques: mapping practices in a global network of universities. Design Science, 10, e7.
Ruiz set up the study, conducted the interviews, a prepared the initial article. Wever provided feedback throughout, participated in some of the interviews, and did the second revision after review.

Paper II
Ruiz JF, Bengtsson, M (to be submitted to Journal) "Into the woods; building trust in teams through scaffolded adventures."
Collaborative work in set-up and execution. Ruiz in the lead of the writing.

Paper III
Bengtsson M, Ruiz, J, Wever, R (2018) THIS IS HOW WE ROLL; A PLAY ON CREATIVE THOUGHT AND THE GENERATION OF NOVELTY. Proceedings of Meaningful Play conference.
Collaborative work in set-up and execution. Co-written by Ruiz and Bengtsson. Authors listed in alpha-betical order.

Paper IV
Ruiz Muñoz, J. F., & Wever, R. (2023). What is your (freaking) Problem? Prototypes for problem exploration on early stages of design. In EKSIG23 proceedings, Milan.
Ruiz set-up the work and wrote vast majority of text. Wever contributed some writing, and feedback throughout.

Other publications not included in the thesis

Wever, R., Ruiz, J., & Bengtsson, M. (2019). Structure versus Meaning: Client-Designer Dynamics around Sustainability Ambitions. In EcoDesign 2019 International Symposium (pp. 217-222).

Ruiz, J. F., & Wever, R. (2022). The Best Of Times, The Worst Of Times; Student Design Team Formation. In DS 117: Proceedings of the 24th International Conference on Engineering and Product Design Education (E&PDE 2022), London South Bank University in London, UK. 8th-9th September 2022.
This paper was selected from the conference proceedings to be expanded and further developed which turned into the appended Paper I .

Wever, R., Ruiz, J., & Bengtsson, M. (2023). An open education platform: Learning our way to mixing life-long learners and regular students. In DS 123: Proceedings of the International Conference on Engineering and Product Design Education (E&PDE 2023).

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Introduction

01

Chapter 1

1. Introduction

Engineering stands at the forefront of modern industry, shaping the world through innovation and technological advancement. Over time, the role of engineers has evolved to better address ever-shifting societal needs, technological advancements, and global challenges. Once primarily focused on the creation of physical structures or machines (Prados et al., 2005; Seely, 1999; Smith, 2024), engineers now find themselves at the forefront of digital transformation, sustainability initiatives, and collaborations with other disciplines (Munir, 2022; Royal Academy of Engineering, 2024; Sheppard et al., 2007). In modern industry, the engineer's role and the skills they need in the development of their profession have drifted from solely being technical skills to also having to include skills in creative problem-solving, design, business and management, cross-disciplinary collaboration, and knowledge integration (Dwan-O'Reilly & Magrini, 2021; May & Strong, 2006; Munir, 2022; Royal Academy of Engineering, 2024; Sheppard et al., 2007).

Technical expertise in natural sciences, mathematics, physics, and engineering principles remains a crucial aspect of engineering problem-solving in creating solutions. However, the types of problems that modern industry and society address have changed, which in turn means that the way in which problems are addressed needs to also change and adapt. The challenges faced by modern problem

solvers are no longer isolated and straightforward, where problem specifications and requirements are clearly defined (Ackoff, 1994; Gharajedaghi, 2011). Instead, problems have become more 'wicked' (Buchanan, 1992), meaning more complex and interconnected, and where problem definition lacks structure and definition and where there is no instant clear path to a satisfactory solution (Buchanan, 1992; Ritchey, 2013; Rittel & Webber, 1973). Furthermore, the intricate landscape of modern industry, with rapid technological evolution, globalization, Interconnected markets, and a societal push for sustainability and social responsibility, further feeds into problems becoming increasingly more complex, interconnected, and ill-defined.

Given the nature of these types of problems, there are no simple formulas or strategies to deal with them (Horn & Weber, 2007; Ritchey, 2013; Rittel & Webber, 1973). While traditional engineering natural science-based linear problem-solving approaches function well in addressing well-defined and isolated problems, they are less suitable for addressing ill-defined and interconnected problems. In a linear problem-solving approach, the sequential, step-by-step process relies on the problem having a clear specification of what the problem is and what a suitable ending point is (Ritchey, 2013; Suoheimo, 2019). To address more ill-defined complex problems, a different problem-solving strategy that utilizes a collaborative and iterative process to explore and frame the different aspects of the problem from multiple perspectives can be utilized (Buchanan, 1992; Horn & Weber, 2007; Ritchey, 2013)

As organizations in modern industries increasingly rely on a growing recognition of the value of diverse perspectives and the importance of combining individual specialists' knowledge within organizations to enhance capabilities (Galloway, 2007; Grant, 1996) for solving complex problems, engineers must then develop a broader skill set

to adapt to this kind of environment. This new set of skills needs to encompass not only the traditional technical know-how that leads to the production of solutions and the previously mentioned ability to define and frame complex problems but also the ability to collaborate and integrate knowledge from diverse perspectives. It also encompasses skills to communicate, negotiate, and come to agreements with individuals across different disciplines (Galloway, 2007; Munir, 2022; Royal Academy of Engineering, 2024; Sheppard et al., 2007)

1.1. Engineering education

Based on the previously described landscape of modern engineering practice, the models and methodologies we use and how students engage in engineering education play a pivotal role in preparing students for the complex and evolving demands of the modern world. Design engineering education is facing several challenges and undergoing significant changes. During the last century, engineering curricula have focused on "engineering science" emphasizing natural science and mathematics as the foundation (Prados et al., 2005; Seely, 1999). However, in the late 1990s and early 2000s, there was a recognition in literature of the need for these educational models to evolve in response to a changing industrial and societal landscape (Ferguson, 1994; Harwood, 2006; Seely, 1999). A recognition that not only places value in the important technical engineering abilities to solve problems but also deals with a growing interdisciplinary complexity of engineering challenges and the need for creativity and innovation in problem-discovery and problem-solving (Dym et al., 2005; Ferguson, 1994).

There was an increasing emphasis on the need for engineering education to include learning experiences that allow students

to construct deep conceptual and experiential knowledge of problem-solving processes that engage in authentic engineering practice(Ferguson, 1994; Harwood, 2006; Seely, 1999). This intends to align existing engineering education with the evolving need for industry adaptability, practical hands-on knowledge, and lifelong learning (Seely, 1999; Harwood, 2006). Furthermore, it provides a foundation and helps to develop in students what Ferguson (1994) and Eisner (2017) describe as a "feel" or developed connoisseurship for problem understanding and solving. The ability to develop a sense of judgment and intuition that spawns from experiential practice and the development of expertise plays a significant role in visualization, spatial reasoning, creativity, multidisciplinary problem-solving, and the integration of diverse forms of knowledge.

1.2. The new Designerly order

To address ill-defined complex problems, organizations have looked to design practices and design problem-solving methods as a new desirable skill for engineering problem-solving (Buchanan, 1992; Horn & Weber, 2007; Suoheimo, 2019). Design problem-solving involves a different approach, which utilizes an iterative approach to identify and define the problem to conceive functional and technically feasible solutions (Buchanan, 1992; Dorst & Cross, 2001).

In the not-very-distant past, the word design was readily associated with the aesthetical appearance of an object or a product. And perhaps still today, in everyday vocabulary, the action "to design" can be associated with changing the appearance of an object to make it "pretty." However, in the organizational and industrial context, the trivialization of design as merely aesthetical has drastically changed over the last 15 to 20 years to encompass an approach to problem-solving and solution generation rooted in how designers approach

problems. The increasing emphasis on design as a problem-solving, sensemaking, and reasoning approach across diverse industries reflects a paradigm shift towards recognizing the multifaceted impact of design in fostering innovation, addressing complex challenges, and enhancing organizational competitiveness (Forrester & IBM, 2018; McKinsey Design et al., 2018).

The significance of design is also evident in the rise of user-centric design methodologies that have made their way into business organizations, governmental organizations, and educational institutions. Of these methodologies, the most well-known is Design Thinking, which has been used in developing innovative products in organizations worldwide, such as Apple, Netflix, IBM, GE, Braun, and Uber, to name a few. The introduction of Design Thinking is often credited to IDEO and Stanford University's Hasso Plattner Institute of Design. It was introduced as a methodological response to addressing complex "wicked" problems in which the approach to creating solutions leveraged user empathy, problem discovery, experimentation, and iteration (Brown, 2008; IDEO, 2018b). However, the methodological tenets found in design thinking are not necessarily new. They can be traced back to the 1970s and 1980s design research literature that explored design as a discipline and attempted to characterize a 'designerly' way of thinking and a 'designerly' way of acting (Archer, 1979; Cross, 1982; Darke, 1979).

Using designerly problem-solving processes and tools in different fields not usually associated with design has become more commonplace. However, its inclusion has also received a fair amount of criticism due to methodological oversimplification and dilution (Cross, 2023; IDEO, 2018b; Johansson-Sköldberg et al., 2013). The criticism of oversimplification is often associated with the sidelining of the active process of iterative exploration, of problem and solution, and the

reasoning pattern of designers (Dorst, 2015) in favor of more palatable and easier-to-consume "easy-to-follow" steps. Johansson-Sköldberg et al. (2013) present a picture of design thinking and its multiple meanings, which they sought to separate into two distinct concepts, 'Designerly Thinking' and Design Thinking. Designerly thinking comes from the academically grounded perspective (Archer, 1979; Cross, 1982; Darke, 1979; Dorst & Cross, 2001; Johansson-Sköldberg et al., 2013), and the more popularized Design Thinking sits closer to the management field. In it, they urge for the importance of Design Thinking to "acquire a scholarly base" (Johansson-Sköldberg et al., 2013, p. 131) for the more critical elements of design cognition to not be lost to a simple set of activities with no designerly knowledge and skill.

With this growing interest in design, and more significantly in the designer's way of solving complex problems from industry, we also can see a growing need and interest from education to integrate and utilize these designerly ways of thinking and acting into educational models (Beligatamulla et al., 2019). Based on this, it is essential for education institutions to respond to industry needs and the demand for designerly problem-solving skills in practice. However, it is vital to do so without falling into the trap of following trends based on methodological oversimplification that is less anchored in the academic design foundation. Instead, it is crucial to provide future problem-solvers with a platform that utilizes design tools for solution creation and design reasoning patterns for problem-solving and sensemaking. Thus, the challenge for educators who utilize models based on design or designerly thinking is to create or utilize structures that support the learning objectives and development of student skills rooted in design reasoning and acting, not merely design tools.

1.3. Why design as an approach to problem-solving?

Design practices and designers distinguish themselves based on their ability to deal with ill-defined, ill-structured, ambiguous, or "wicked" problems where a single optimal solution may not exist (Buchanan, 1992; Rittel & Webber, 1973; Simon, 1973). Rather, designers explore, define, and redefine problems to provide sufficient definitions and structure to propose acceptable and appropriate solutions rather than strictly optimal ones (Archer, 1979; Buchanan, 1992; Cross, 1982; Simon, 1973; Voss, 2005). To address these types of problems, design practitioners engage in a simultaneous exploration of both problem and solution spaces, constantly refining their understanding through iteration (Dorst & Cross, 2001). This strongly emphasizes developing temporary solutions or prototypes to provide structure and definition to the problem (Archer, 1979; Cross, 1982; Dorst & Cross, 2001; Lawson, 2006).

The ability of designers to oscillate between problems and solutions, rooted in solution finding, is evident when comparing their problem-solving approach to that of the natural sciences. From a natural science perspective, the problem-solving process aims to study problems, seeking a single optimal solution by discovering their governing rules (Cross, 1982; D. H. Jonassen, 2000; Lawson, 2006). In contrast, designers use a temporary solution to understand the problem space and potential solutions (Cross, 1982; Darke, 1979). Designers refine their understanding through iterative testing and propose new, better-fitting solutions rules (Cross, 1982; D. H. Jonassen, 2000; Lawson, 2006; Darke, 1979). This action of an ongoing iterative exploration of the problem through the production of solutions allows designers to provide, with each iteration, incremental definition and structure to the problem where new elements are proved in the light of temporary solutions.

1.4. Cross-disciplinary boundaries.

As previously discussed, in the landscape of modern industry in which engineering professionals deal with complex and ill-defined problems, the ability to communicate, negotiate, collaborate, and integrate knowledge across disciplinary boundaries becomes relevant (Galloway, 2007; Munir, 2022). Skills that involve effective communication and negotiation can bridge the gaps between different disciplines to tackle problems that transcend traditional disciplinary boundaries. As education continues to evolve, its emphasis on multidisciplinary collaboration becomes increasingly vital as it provides students with the necessary skills and preparation to cooperate with individuals from other disciplines.

Higher education involves distinct fields focusing on specific areas of study with unique bodies of knowledge and methodologies. This division promotes specialization, contributing to knowledge advancement and skilled professional development (Schofer & Meyer, 2005). While this specialization has been crucial for societal progress and economic development, it also creates boundaries between professionals in separate fields (Schofer & Meyer, 2005). The increasing specialization and compartmentalization of knowledge highlight the importance of understanding and navigating these boundaries in multidisciplinary collaboration and knowledge sharing as they define the scope and limitations of shared understanding and communication (Akkerman & Bakker, 2011).

Boundaries are ”real or imagined lines marking the edge or limit of something” (Cambridge Dictionary, 2024). In social interactions, boundaries can separate different groups, creating limits in understanding. These conceptual divisions are not necessarily only problematic stoppages in interaction, but instead, they represent

points of intersection where different knowledge groups will need to come together (Carlile, 2002; Star & Griesemer, 1989; Wenger, 2000). Within the context of these disciplinary divisions or boundaries, we find boundary objects described as artifacts that function as communication bridges to find common understanding and facilitate interaction between groups (Star & Griesemer, 1989). In the context of education, Wenger's (2000) description of boundary objects highlights the role of these objects as bridges that mediate, translate, and facilitate a common understanding among students and educators.

So, in the face of an interdisciplinary and multidisciplinary approach to education, it is important for educators to utilize processes, methods, or objects that facilitate the crossing of disciplinary boundaries. From the educators' perspective, this means examining, understanding, and purposefully planning for the use of boundary-crossing elements that help students from different disciplines to collaborate effectively. This is both to facilitate student collaboration and, as part of the learning objectives, to provide practical experience to students on how to utilize these tools in their future professional careers.

1.5. Purpose, Aim, and Research Questions.

In this thesis and the studies within it, I seek to explore the early part of the designerly problem-solving process. The focus, however, is not on solving problems and creating solutions but on exploring, understanding, and framing complex problems. How this exploration of the problem is done in a multidisciplinary environment where collaboration across disciplinary boundaries and integration of knowledge from multiple perspectives occur. And how, as teachers of multidisciplinary teams, we educate students to do so while

simultaneously crossing disciplinary boundaries, not only with students but also with other stakeholders in the educational process. This framing of the thesis then leads to an aim or expected goal of the thesis, the general purpose to do so, and the research questions that will guide the thesis exploration process:

Purpose:

The overarching purpose of my research project is related to the central question of how might we better prepare our students, as future problem-solvers, to deal with, explore, define, and frame complex problems in a changing environment that includes multidisciplinarity, team diversity, remote collaboration, new technological development, among others? More specifically, how might we utilize design boundary objects such as artifacts, processes, and discourses to aid in exploring problems while simultaneously serving as boundary-crossing bridges in multidisciplinary collaboration? The main purpose is to generate design knowledge that is practical, readily actionable, and usable by other educators in the context of designerly education.

Aim:

The aim of my research project can be divided into two parts. Firstly, to study practices and processes utilized by educators in problem-based designerly education to understand the underlying mechanisms and theoretical underpinnings in the process of problem exploration in multidisciplinary education. To discuss and present how this mechanism at play can enhance designerly problem-based educational methods for problem exploration and framing. Secondly, to explore and discuss the same processes and methods, and the tools and activities within them, in the context of multidisciplinary education that help to bridge boundaries between participants from different disciplines. The intent is to explore how using designerly objects as

boundary objects facilitates communication and collaboration of students, educators, and other participants and how those objects enhance our educational practices.

Based on this, the following research questions were formulated.

> ***RQ1:*** How might we as educators provide our students with problem framings in the form of design briefs that provide them with a starting point framed in a way that fosters problem exploration? How can the brief and the process of creating it serve as bridges and boundary-crossing artifacts that aid in communication, negotiation, and collaboration between students, teachers, and other stakeholders of their educational process?
>
> ***RQ2:*** How might we use existing design methods, such as ideation and prototyping, to help students better define, understand, and reframe problems through temporary discardable solutions? How does using ideation and prototyping methods that focus on problem exploration serve as boundary object artifacts help facilitate these processes and bridge disciplinary boundaries between students, teachers, and other stakeholders?
>
> ***RQ3:*** How might we create teams and build team dynamics that represent real-world diverse and multidisciplinary team situations and are conducive to achieving the learning objectives? How might we construct and use boundary object processes to help bridge team creation between educators and as processes to develop team dynamics?

1.6. Thesis outline

Throughout this thesis, I will present the different studies conducted during my doctoral process, their theoretical connections, methodological development, and analysis. To do so, it is important to consider the context of the educational platform in which these studies were conducted and how they fit within that logical structure. Thus, I will not utilize a chronological timeline of the order in which the papers were produced during my doctoral studies. Rather, for the analysis of the papers and how they fit together as a thesis, I will be presenting them according to how they fit within the chronology of the project process followed by the students and educators in a project-based learning educational platform at Linköping University and as part of a network of international student projects.

1.6.1. Research context

In 2016, Linköping University started the Design Thinking and Multidisciplinary Projects (DaMD) course based on Stanford University's ME310 Global Engineering Design Innovation course. The projects and the course are part of an international collaboration with Stanford's ME310 and the SUGAR global network (a spin-off from ME310). The course is set up as project-based and mission-driven, in which students learn theory and methods for problem-solving and product development (Wever et al., 2023) Bengtsson et al. 2024). Students engage in a hands-on experience of solving a 9-month project provided by a sponsoring company that starts from a problem brief to a proof-of-concept solution. The projects are done in multidisciplinary groups of students in internationally distributed teams from engineering, design, and strategy master programs (Wever et al., 2023).

The course process involves a combination of design thinking methodology and tools, specific missions, and prototyping assignments that follow the Stanford ME310 process, which is meant to guide students through a divergence and convergence exploration of problems and solutions (figure 1). Teams regularly present and document their progress throughout project development, divided into three quarters with 2 to 4 missions each. Informal meetings during missions provide opportunities for feedback, and at the end of each quarter, formal presentations that encapsulate the team's progress and process. The project culminates in a final presentation at a design fair, featuring physical prototypes, short presentations, and graphic materials documenting the final result and the iterative process.

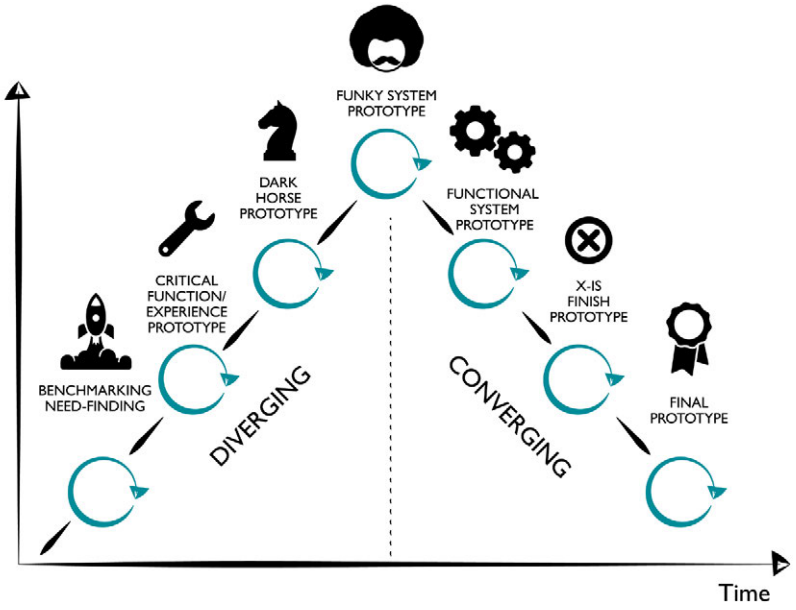


Figure 1. ME310 course process and missions

Nevertheless, how the projects and the course develop within the timeline presented above relates mainly to how students might experience the course. It is important to point out that from the teacher's perspective, there is a 'Planning' sequence of events in which certain preparation needs to be done before the project begins, and that plays an important role in achieving the educational goals. However, the initial 'Planning' is never seen from the student's perspective.

From the teaching perspective, the progression that follows (figure 2) is divided into three distinctive phases base. Firstly, a Project Planning Phase in which the Teaching team prepares the necessary elements of the projects. Once the projects have started, the second phase, in which the project is in the early stages of development, can be characterized as the Divergence Phase of the design process, in which the focus is problem exploration and problem framing. Finally, a Convergence Phase that focuses on solution development and that resembles the traditional design engineering process of product development in the traditional Product development framework (Ulrich et al., 2008).

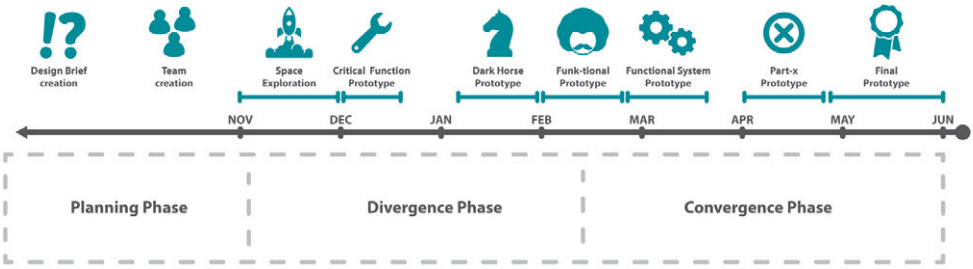


Figure 2. Course process including phases.

In terms of the research studies I conducted, my experiences in teaching design product development have helped me recognize an area of a common struggle that most teams of students had in the

early stages of their projects. As designers and engineers, students are generally well-equipped to create and build solutions capable of performing as intended. However, during the early stages of the process, most teams of students struggle to identify and frame interesting problems to produce solutions and in the process of doing so they also struggle to collaborate and communicate with other students from other disciplines. Based on this and given that this particular thesis focuses on the exploration and framing of ill-define problems and boundary crossing in multidisciplinary collaboration. The studies conducted are as follows (Figure 3): Licentiate Thesis on design briefs and problem formulation. Paper I on team formation and Student assessment methods. Paper II on Team Building and Trust Building. Paper III on randomness and generation of novelty through an ideation technique. And Paper IV on prototyping for problem exploration.

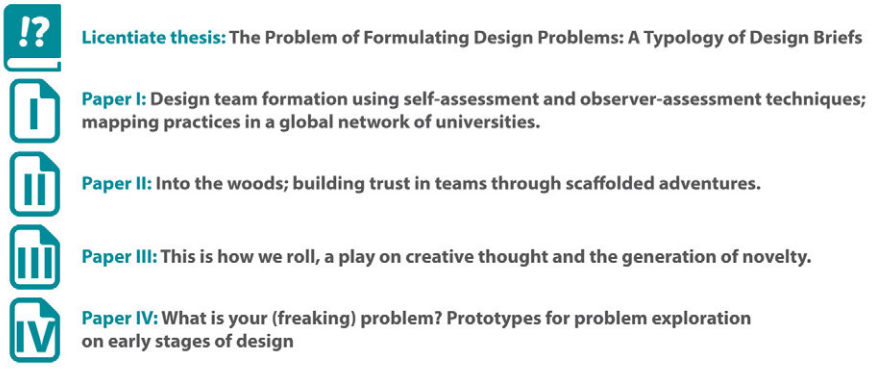


Figure 3. Appended studies

The papers presented here are mainly placed in the early stages of the projects (figure 4). The Licentiate thesis, Paper I, and Paper II are within the Project Planning Phase and follow the order of Problem definition and brief creation, Team creation, and team building, which occurs before the projects have commenced. Paper III and Paper IV are within the Divergence phase, in which teams explore

the design challenge and open up their understanding of how tools such as ideation and prototyping can aid in framing and reframing the problem.

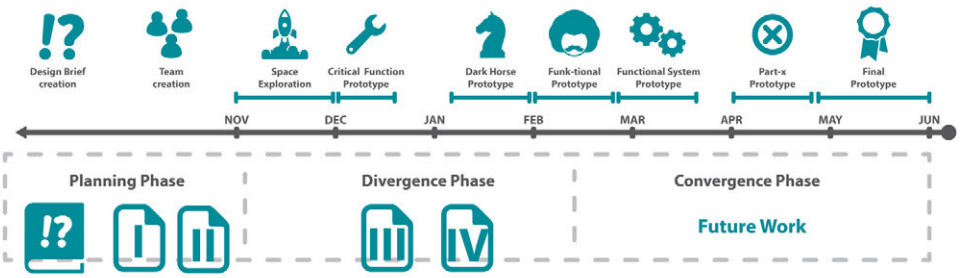


Figure 4. Studies of this thesis placed in timeline.

1.6.2. How the research was conducted.

As previously mentioned, my doctoral research has been conducted over the central platform of the Design Thinking and Multidisciplinary Projects course. This means my research and the publications presented here are intricately connected to a practical setting of a live and ongoing educational model. Based on this, I conducted research associated with the practices and methods developed through years of practical experience by different educators from the multiple teaching teams involved with the global network. While simultaneously developing our methods and practices on the Linköping University teaching team.

In the licentiate thesis study, I set out to research design brief as the starting point of the projects and the first point of contact that the participating students had with the problem that they would then solve. The intention of the study was to examine different starting points and what these starting points mean for the problem-solving process that follows. To do so, I studied and codified 118 design briefs from the ME310 course at Stanford University based on the

different attributes that the briefs possessed. With this, a typology of design briefs was created, and an analysis of what the different types of starting points could mean for the problem-solving process. Furthermore, a brief creation talking tool was created to serve as a bridge between educators and commissioning organizations to better align their briefs with the desired outcomes.

In Paper I on team formation, we set out to study the formation of student teams in project-based education settings. To do so, I conducted semi-structured interviews with educators from different universities in the SUGAR network of projects. In the interviews, several aspects of the team formation process were discussed, such as different attributes that teams and students need for teams to work cohesively, how these attributes are observed and assessed by educators, and how they are combined to create teams. Based on this, a thematic analysis discussed several aspects of the methods and practices used by the educators and teacher expertise and experiential knowledge in the assessment process.

In Papers II, III, and IV, I utilized action research, which emphasizes a participatory approach, to address practical problems (Altrichter et al., 2002; Säfsten & Gustavsson, 2020) and research through design, which focuses on tangible artifacts and reflecting on them to generate knowledge (Findeli, 2010). In Paper II, we studied a trust-building exercise that utilizes scaffolded activities on an adventure-based exercise utilized in a master's program at Linköping University to develop trust among new students through swift trust and shared experiences. We used trust-building literature and the different exercises designed and utilized in action to examine how trust is built through engaging students in collaborative and "high-stakes" tasks. In Paper III, we developed an ideation game that was used in the DaMD projects, which utilized randomness to generate novel ideas

and explore different problem framings to avoid design fixation. In the paper, and utilizing novelty generation literature, we explore how the introduction of randomness can aid in avoiding the negative effects of design fixation and functional fixedness in creating new problem frames where new solutions can be proposed. Finally, in Paper IV, we studied early-stage prototyping to explore and identify what it means to build temporary and discardable prototypes with the sole intention of exploring problems. For this, we studied different prototypes created in the frame of the DaMD and SUGAR projects, which contrasted with the specific missions developed for these courses, served a different purpose than testing possible solutions, and aimed at uncovering new information about the problem space. To do so, we used prototyping literature and reflection in action literature to explore how prototypes can be used to reflect, not on their effectiveness as a solution, but to prove questions regarding the problem at hand.

1.6.3. Thesis structure

This thesis is divided into four chapters. They are as follows:

Chapter 2. Theoretical Frame

This chapter will introduce the theoretical foundation that was used in this thesis. The literature introduced in this chapter will be on design science, designerly thinking, and Design Education. More specifically this chapter will examine the evolution of design engineering education and the push for designerly ways of thinking and doing to be included in engineering education curriculum through educational models such as problem-based learning and challenge-based learning. Furthermore, this chapter will examine boundary objects and their function within design and design education.

Chapter 3. Methodology

This chapter will introduce the research approach used to create this thesis. This will include the general research approach utilized through my doctoral education in the development of multiple studies, as well as my philosophical and epistemological view of design research. The specific research methodology and its use in the individual studies will also be explained.

Chapter 4. Analysis and contributions

This chapter will describe the analysis and contribution of the different studies conducted in the context of the educational model of design thinking and PBL/CBL and how they contribute to understanding and developing said educational model. Moreover, an analysis of how the different boundary objects of design described in the different studies contribute to achieving educational goals under the educational model. This will be done using the educational model process and timeline as a background to how the different studies contribute to the ongoing educational process.

Chapter 5. Conclusions

This chapter will present the conclusion and answers to the research questions based on what has been discussed throughout the thesis. Also, a discussion of limitations and implications will be presented, concluding with a discussion of future work.

02

Theoretical Frame

Chapter 2

2. Theoretical frame

2.1. Design (&) engineering education.

Historically, Engineering can be traced back to ancient civilizations around the globe, where skilled craftsmen and artisans developed techniques for constructing monumental structures using simple engineering (Smith, 2024). However, it wasn't until the Industrial Revolution in the 18th century that engineering began to emerge as a distinct field of study (M. Cardoso & Chanin, 2022). The establishment of formal engineering education institutions, such as the École Polytechnique in France in the late 18th century and the Massachusetts Institute of Technology (MIT) in the early 19th century, further solidified engineering as an academic discipline (M. Cardoso & Chanin, 2022).

During the 19th century, the practice of engineering was still strongly associated with hands-on practical experience, with the primary goal of education being the enhancement of practical application (Prados et al., 2005; Seely, 1999). However, during the late 19th century and early 20th century, an interesting shift with a simple yet important recognition that students of engineering needed to transcend just knowing how (practical) things work to also know why (underlying principles) things worked (Seely, 1999). In the early 20th century, engineering education shifted towards a stronger mathematical and

theoretical foundation. Influenced by the development of scientific-based approaches, favored in French and English Engineering schools and adopted in the U.S. with the arrival of prominent engineering theorists from Europe (Prados et al., 2005; Seely, 1999). This period marked a pivotal moment in the development of engineering education, shaping the way engineering principles were taught and understood as it drifted towards an 'academization' of engineering education. Or a drift from academic institutions towards engineering education that was more preoccupied with developing scientific-based theoretical models than their practical application (Harwood, 2006).

However, this led to growing recognition during the early 1990s and early 2000s that engineering students who transitioned to professional practice lacked important hands-on practical skills essential to engineering practice (Harwood, 2006; Prados et al., 2005; Seely, 1999). Key among said skills are team-oriented and multidisciplinary collaboration (Seely, 1999), but mainly and more importantly, problem-solving and problem-understanding skills and abilities that sit to design practice and experience (Ferguson, 1994; Harwood, 2006; Seely, 1999).

As engineering education continued to advance into the 21st century, the need for problem-solving abilities and design experience became increasingly recognized as essential capabilities for engineers (Galloway, 2007; May & Strong, 2006; Munir, 2022; Sheppard et al., 2007). This recognition led to a rethinking of engineering education, emphasizing the importance of incorporating design capabilities into engineering curricula (Ferguson, 1994; Harwood, 2006; Seely, 1999). However, as pointed out in the literature, it is important not to over-correct but rather to find a balance between the how-to-do and why-do-it of scientific-technical abilities to meet designerly problem

understanding (Jamison et al., 2014) with further transversal and soft non-field specific skills (Galloway, 2007; Jalali et al., 2022; Munir, 2022).

To achieve this, several educational models that place a greater focus on design-engineering problem-solving while harnessing technical know-how have become more common in design engineering education (Guerra, Ulseth, & Kolmos, 2017; Guerra, Ulseth, Jonhson, et al., 2017). Methodologies such as Project-based learning (PjBL), Problem-based learning (PBL), Conceive Design Implement Operate (CDIO), and Challenge-based learning (CBL) utilize problem-solving activities such as collaborative projects to engage students in learning activities.

2.1.1. PjBL, PBL, & CBL

In recent years, these approaches for active education have regained attention in engineering education (Sukackė et al., 2022) despite existing for a long time in other educational contexts. Each of these approaches refers to a specific model of the educational process in the classroom. Yet, there is certain confusion regarding the use of the terms based on overlapping aspects that the models share. PjBL, PBL, and CBL share multiple central aspects and tenants that emphasize desirable learning outcomes for design and engineering education. They are rooted in the idea of providing students with real-world problems or challenges, fostering deeper understanding through active participation (LaLopa & McDonald, 2002; Stepien & Gallagher, 1993). Students then engage in self-directed learning, conducting research, integrating theory and practice, and applying knowledge and skills to develop a viable solution to the problem. They emphasize the process of problem-solving and encourage students to explore multiple perspectives and potential solutions ((Sukackė et al.,

2022)). Their effectiveness in improving students' creativity has been highlighted in the literature, making it a valuable model for enhancing design learning in engineering education (Cahyani, 2021); however, as mentioned before, it is important to understand their differences and how they relate to each other.

What are they?

Project-Based Learning (PjBL): PjBL is an instructional methodology centered on student-driven projects that culminate with the creation of artifacts or end products as outcomes of the process (Kokotsaki et al., 2016). PjBL traces its origins back to the early 1930s in connection to John Dewey and his educational philosophy, in which learning can naturally be achieved through the doing or execution of a project or task that demands thinking in order to be achieved (Dewey, 1930). From the 1970's, it gained prominence in engineering education (Sukackė et al., 2022). In PjBL, students embark on a process to solve a problem or challenge, often within a context of interdisciplinarity (Kokotsaki et al., 2016; Sukackė et al., 2022). In this approach, projects are designed to simulate real-world scenarios and require students to apply acquired knowledge and skills to solve problems. The teacher acts as a facilitator, guiding and supporting students while allowing them autonomy to make decisions and drive their learning journey (Helle et al., 2006; Kokotsaki et al., 2016; Sukackė et al., 2022).

Problem-Based Learning (PBL): The concept of Problem-Based Learning (PBL) originated at McMaster University in the late 1960s in health education and quickly expanded in the 1970s to many other institutions and fields of study in science, social sciences, and humanities (Edström & Kolmos, 2014; Kolmos et al., 2009). It is a student-centered pedagogical approach that focuses on using real-world problems as a context for learning, aiming to develop

experiential, hands-on learning (Edström & Kolmos, 2014; Kolmos et al., 2009). PBL is designed to help students develop problem-solving skills, as well as other transversal skills such as communication, collaboration, and reflective thinking (Edström & Kolmos, 2014).

In terms of the use and application of PBL in different situations in a classroom, Kolmos et al. (2009) present us with a separation of approaches to PBL that can be read as extremes within a spectrum. On one side, a discipline-specific PBL model based on narrow, disciplined-specific learning goals and a teacher-controlled approach where learners are presented with well-defined problems with a known/knowable solution and where the main objective is to develop field-specific skills. On the other hand, a more innovative PBL with a learner-centered focused approach with Ill-defined problems where the solution to the problem is unknown to the instructor and where the educational goal is the development and self-construction of interdisciplinary knowledge (figure 5).

Challenge-Based Learning (CBL): CBL is an educational approach that frames learning around solving complex and ill-defined real-world problems or challenges (Sukackė et al., 2022). CBL projects typically involve interdisciplinary collaboration, leveraging diverse perspectives and expertise to tackle multifaceted problems. In comparison to both PjBL and PBL, CBL is a much younger educational concept or approach, and its development can be traced to an initiative by Apple, Inc. in 2008 (Nichols & Cator, 2008). CBL can also trace its roots to PBL as its precursor. However, researchers have sought to distinguish CBL from PBL based on the type of problem they address, with PBL framed as a more domain-specific problem and CBL as the more open and innovative problem. However, based on the previously presented distinction by Kolmos et al. (2009), we can position CBL with PBL at the end of the spectrum where a more innovative learner-

	Discipline-specific PBL	Innovative PBL
Objectives and knowledge	Traditional discipline objectives Disciplinary knowledge	Project-based objectives Interdisciplinary knowledge
Type of problems	Narrow Well-defined problems	Open Ill-defined problems
Students' learning	No supporting courses Acquisition of knowledge Collaboration for individual learning	Supporting courses Construction of knowledge Collaboration for innovation
Academic staff and facilitation	Discipline specific training Teacher-controlled supervision	More training required Facilitator/ process guide
Space and organisation	Administration from traditional course and lecture-based curriculum on lecture rooms	Administration support for PBL curriculum & Physical space to facilitate teamwork
Assessment and evaluation	Individual assessment Summative course evaluations	Group assessment Formative evaluation

Figure 5. Spectrum of PBL from discipline-specific to innovative (Based on Kolmos et al. 2009)

centered focused approach to Ill-defined problems. Yet the true distinction of CBL lies in its focus on addressing bigger societal and environmental problems (Nichols & Cator, 2008) with solutions that are "environmentally, socially and economically sustainable" (Kohn Rådberg et al., 2020, p. 22).

Overlapping and differences

As previously discussed and based on the definitions presented above, PjBL, PBL, and CBL share similarities and overlaps in their approach to education. All three methodologies prioritize hands-on, experiential learning where students actively engage with authentic problems or projects. All three emphasize the application of knowledge and skills to real-world problems/challenges, fostering

connections between classroom learning and practical experiences. Finally, in all three models, students collaborate in teams, drawing on diverse perspectives and expertise to explore problems, generate solutions, and create solutions (Edström & Kolmos, 2014; Kohn Rådberg et al., 2020; Kolmos et al., 2009; Malmqvist et al., 2015; Sukackė et al., 2022)

However, where there are similarities, there are also differences that separate PjBL, PBL, and CBL. Firstly, based on the type of problematic situation that each aims to address, PjBL typically takes as a start a project or task designed by the teacher, already containing a certain level of structure (Sukacke et al., 2022). PBL can be positioned within the spectrum on the one side with discipline-specific PBL, which is very similar to PjBL with a well-structured problem in which the instructor knows the possible answer. On the other end of the spectrum, innovative PBL begins with a problem scenario that is an open-ended challenge and where part of the process is to frame the problem being solved (Kolmos et al., 2009). And with CBL starting in a very similar position as innovative PBL with an open-ended challenge that centers around significant global challenges, with issues relevant to society (Kohn Rådberg et al, 2022).

Secondly, based on the type of collaboration, while PjBL and discipline-specific PBL projects focused on problem-solving that is discipline-specific, there is typically collaboration with participants from a single discipline (Sukacke et al., 2022; Edström & Kolmos, 2014; Kolmos et al., 2009). Innovative PBL and CBL address challenges that pertain to several disciplines and that transcend traditional academic boundaries. An important element of the problem-solving process is to work in settings of multidisciplinary collaboration, which might also involve internal and external stakeholders in the project (Sukacke et al., 2022; Kohn Rådberg et al., 2022; Malmqvist et al., 2015).

Finally, we can differentiate them based on the actions that students engage in during the design process. With PjBL and discipline-specific PBL as more directed to the mobilization of knowledge (Sukacke et al, 2022) meaning the use of discipline-specific knowledge to the creation of solutions. And with innovative PBL and CBL as more oriented towards the acquisition of knowledge (Sukacke et al, 2022) in which the participants develop and constructure knowledge of the problem and of the solving of the processs.

2.1.2. PBL in Education.

This approach, despite existing for a long time, has recently gained attention in engineering education and shares multiple central aspects that emphasize learning outcomes of design and engineering fundamentals (Kolmos et al., 2009). They are rooted in the idea of providing students with real-world problems or challenges, and that can be used in multiple contexts, from secondary education to higher education and in multiple fields of knowledge (Edström & Kolmos, 2014; LaLopa & McDonald, 2002; Stepien & Gallagher, 1993). However, the wide range of adoption and academic development of PBL as an education model also means that there is no clear, standardized, or unified model for PBL (Kolmos et al., 2009). However, researchers such as Kolmos et al. (2009) and Savin-Baden (2000 & 2007) have sought to provide structure to another wise, very spread concept. Firstly, Kolmos et al. (2009) have defined the common learning principles (figure 6) that define PBL (Edström & Kolmos, 2014; Kolmos et al., 2009). At the same time, Savin-Baden (2000 & 2007) goes further to characterize PBL based on six different dimensions to propose different types of models of PBL (figure 7).

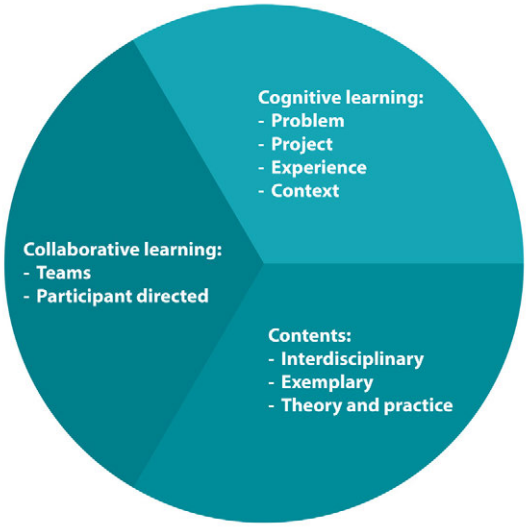


Figure 6. PBL common learning principles (Adapted from Kolmos et al. 2009)

The implementation of PBL in engineering education aligns with the broader trend towards increasing the design component in engineering curricula, aiming to better prepare graduates for engineering practice (Dutson et al., 1997). PBL not only enhances students' learning motivation and problem-solving abilities but also provides opportunities for them to develop deep learning through engagement with real and relevant problems (LaLopa & McDonald, 2002; Stepien & Gallagher, 1993). Moreover, in the case where team-based PBL is also a component of the pedagogical methodology, it provides students not only with opportunities to engage in learning-by-doing but also allows for the development of other interpersonal skills, which are essential in design engineering practice (Dym et al., 2005). PBL's effectiveness in improving students' creativity has been highlighted in the literature, making it a valuable model for enhancing design learning in engineering education (Cahyani, 2021).

Model I **PBL for epistemological competence**	• Knowledge: Propositional knowledge • Learning: use of propositional knowledge • Problem scenario: limited problems with known solutions • Students: are receivers of propositional knowledge • Facilitators: guide to understand the correct knowledge • Assessment: test of knowledge according to objectives
Model II **PBL for professional action**	• Knowledge: know-how • Learning: skills for workplace • Problem scenario: real life • Students: Solve real problems in order to take practical actio • Facilitators: demonstrator of practical skills • Assessment: Skills for the workplace and supporting knowledge
Model III **PBL for interdisciplinary understanding**	• Knowledge: a cross between know-how and know-what • Learning: knowledge and skills across discipline boundaries • Problem scenario: centred around knowledge with action • Students: integrators across boundaries • Facilitators: coordinator of skills and knowledge across boundaries • Assessment: skills and knowledge in a relevant context.
Model IV **PBL for trans-disciplinary learning**	• Knowledge: to understand existing boundaries • Learning: critical thinking and understanding discipline borders • Problem scenario: dilemmas • Students: independent thinkers with a critical stance • Facilitators: flexible • Assessment: To demonstrate integration across disciplines
Model V **PBL for critical contestability**	• Knowledge: contingent, contextual and constructed • Learning: interrogation of frameworks • Problem scenario: multidimensional, offering alternative ways of understanding • Students: explorers of underlying structures and beliefs and developing new hypotheses and knowledge • Facilitators: commentator, challenger and decoder of cultures, disciplines and traditions • Assessment: open-ended and flexible

Figure 7. Models of PBL proposed by Savin-Baden (2000 & 2007) and adapted by Kolmos et al. 2009)

However, these pedagogical approaches come with limitations of their own. Given the variability in which PBL can be implemented in classrooms, multiple aspects need to be taken into consideration before implementing PBL practices. Aspects such as different levels of instructor involvement and facilitations, instructor expertise, problem formulation, participants involvement, and level of interest, among others. This in turn means that the fulfilment of the educational goals can oscillate between positive and negative depending on its implementation (Prince & Felder, 2006).

2.1.3. CDIO Framework

The CDIO framework, which stands for Conceive, Design, Implement, and Operate, is a comprehensive approach to engineering education that emphasizes the real-world application of engineering principles and skills. It was developed as an alternative to traditional linear engineering curricula with the intention of promoting active learning and problem-solving skills in engineering education. While it shares certain core characteristics of PBL, the main separator of the two approaches centers on the unstructured and ununified nature of PBL, while CDIO is a framework with a set of standardized guidelines. CDIO provides a structured method for curriculum design, assessment, and implementation, with a focus on producing engineers who are capable of effectively conceiving, designing, implementing, and operating complex systems and products (Crawley et al., 2011). The framework is designed to integrate practical knowledge and skills into engineering education, emphasizing the importance of hands-on experience and project-based learning (Gunnarsson & Swartz, 2022).

The CDIO framework has been implemented in various engineering courses, including mechatronics and computer numerical control application courses, with positive outcomes in terms of student-centered learning and the improvement of engineering education (Malmqvist, 2012). Moreover, the CDIO framework has been recognized as an innovative educational model that fosters the development of engineering students by integrating a comprehensive set of skills and knowledge with disciplinary expertise (Torsakul et al., 2019).

Furthermore, both PBL and the CDIO framework have been widely recognized for their effectiveness in engineering education, emphasizing the real-world application of engineering principles

and skills. However, it is important to acknowledge that a limitation associated with these approaches is the potential challenge of fully integrating transversal skills, such as communication skills, interpersonal skills, and intrapersonal skills, among others (Jalali et al., 2022). While the educational approaches provide a structured approach to engineering education, there may be a need for additional emphasis on these transversal skills to ensure that students are well-equipped for professional engineering work.

2.2. To Design or Not to Design

Since the late 14th century, the word design has been utilized to describe both the process of making and shaping things as well as to do so in a planned way. These early uses of the word already point to many of the current definitions of design as the purposeful process of creating artifacts that help shape the world around us. Design as a field of study is multifaceted and has been approached and investigated from many perspectives, which also lends itself to a multitude of definitions of what design is and what it concerns itself with.

In his seminal book, The Sciences of the Artificial, Herbert Simon presents us with the view that we live in an artificial world rather than a natural one (You & Hands, 2019) as it is a world affected, shaped, and designed by humans with systems and artifacts to serve a specific purpose. Based on this, Simon proposes a definition of design, or moreover, of designers, "Everyone design who devises courses of action aimed at changing the existing situation into preferred ones" (Simon, 1969, p. 55). This way of understanding design remains, at its core, deeply connected to the initial definition of design as the purposeful action of creating and shaping.

Through the many years, both before and since Simon's Sciences of the Artificial, many design scholars have sought to enrich our

understanding of what design is that goes beyond the simple act of creating but also explores the actions and processes that are involved in its practice. Furthermore, authors such as Schön (1983) and Cross (1982) have sought to challenge the notion of design being a science or a form of artistry to design being its own form of "designerly ways of knowing, thinking and acting" (Cross, 2001, p. 55). Based on this, Design has been defined as the process of creating and shaping artifacts, systems, or environments to fulfill specific purposes or solve problems (Archer, 1979; Cross, 1982; Dorst & Cross, 2001; Simon, 1969). This process involves the deliberate exploration of problems within the given context via an iterative process of generation and evaluation of potential solutions. This often involves understanding user needs, conducting research, and considering constraints and trade-offs to develop satisfactory solutions (Archer, 1979; Cross, 1982; Dorst & Cross, 2001; Simon, 1969).

Yet, if we examine the current landscape of design practices contrasted with Simon's definition of "everyone who designs," we encounter a multitude of disciplines in which design plays a critical role. The word design or the verb to design is often utilized in relation to multiple disciplines such as Engineering, Management, Arts, and Computer Science, among others. Given the versatility of the term design, it is important to distinguish what makes design, designerly actions, and design practitioners distinguishable from those of other disciplines.

2.2.1. What does it mean to design?

Design distinguishes itself from other disciplines by several important factors. Design can be distinguished by the type of problems it aims to solve and the methods employed by designers to tackle these problems (Buchanan, 1992; Cross, 1982, 2001; Simon, 1973). From design literature, we can observe that beyond the multitude of definitions of

design that can be found. Several authors seem to agree that design engages in solving problems that are poorly defined or ambiguous in nature and where a single optimal solution might not be reached. This means that knowledge of the problem, variables and limitations, and criteria for a solution or goal are not fully known, and where the information available might be conflicting and confusing (Archer, 1979; Buchanan, 1992; Cross, 1982; Simon, 1973; Voss, 2005). These types of problems are referred to in design literature as Ill-define or ill-structured problems (Buchanan, 1992; Cross, 1982; Newell & Simon, 1972). As a result of their lack of definition, an important element of the design practice is to continuously explore, define, and redefine the problem being solved to allow for the creation of appropriate and acceptable solutions rather than optimal ones.

Based on the type of problems addressed by design, another factor that distinguishes design from other disciplines relates to the process in which said problems are explored and solved. Designers engage in a simultaneous co-exploration of both problem and solution spaces, constantly refining and iterating their understanding of the problem as well as possible designs that achieve a desirable outcome (Dorst & Cross, 2001). To do so, designers also place a strong emphasis on an iterative process of exploring the problem by developing temporary solutions or prototypes that allow them to provide structure or definition to the problem (Archer, 1979; Cross, 1982; Dorst & Cross, 2001; Lawson, 2006).

Furthermore, based on this need to "open up" and explore all possible variables of a given problem and how they connect to multiple stakeholders, the design practice involves a holistic and interdisciplinary approach that considers many different facets of the problem. In order to achieve this, designers draw upon knowledge and skills from various fields, such as engineering, art, management,

psychology, and social sciences, among others, to inform their process (Murphy et al., 2015). This interdisciplinary perspective allows designers to consider multiple factors and perspectives when developing solutions that lead to comprehensive outcomes based on a deep exploration of the problem at hand and the ability to propose solutions that also draw elements from the various fields mentioned above. This, in turn, also provides design and design practitioners who inhabit specific disciplines the possibility to explore and collaborate in problem-solving processes from multiple domains.

This interdisciplinary approach means that designers integrate multiple aspects of functionality and usability as well as aesthetic considerations (Murphy et al., 2015). Design is characterized by its focus on the user and the consideration of their needs, desires, and behaviors. Designers aim to create products or solutions that not only solve problems but also engage aspects of user needs and desirability for a given solution while also pairing these discoveries with technical feasibility as well as business viability ((IDEO, 2018a)). This focus on the human-centered aspect of design distinguishes it from disciplines of engineering and or management that may prioritize technical and technological feasibility or business viability, respectably, over a more user-centric approach. Designers conduct user research and engage in empathetic design practices to gain insights into user preferences and behaviors (Brown, 2008; Cross, 1997; Lawson, 2006).

These different perspectives on design point to design not being limited to the creation of physical artifacts but extending to problem-solving, decision-making, and shaping the environment. It involves iterative processes, exploration of possibilities, and consideration of contexts. Designers engage in divergent thinking to generate multiple ideas and converge on the best possible solution through experimentation and iteration. Design also involves empathy and

understanding the needs and perspectives of users. Additionally, design can be approached systematically, with a focus on problem-solving and decision-making.

2.2.2. Designerly ways of knowing and doing

With the express intent of establishing design as its own discipline, several researchers, among which we can find Archer (1979), Nadler (1980), and Cross (1982), along with researchers from the Royal College of Arts, sought to characterize the nature of design as the 'third culture' of humane knowledge along with natural sciences and humanities. Among the main outcomes of their effort, we can find the characterization of the phenomenon study by design: that of man-made objects, the methods by which it is examined: modeling, pattern-formation, and synthesis, and the values of design: practicality, ingenuity, and empathy (Cross, 1982).

With this characterization of the design practice also came a characterization of how designers behaved, with the intent to present that design had its own distinct set of things to know and a distinct designerly way of knowing them (Cross, 1982). As presented before, design is characterized by tackling ill-defined problems. Yet, Archer (1979) argued that most people face this kind of problem in life, and the uniqueness of the designer's way of acting lies in the ability to "oscillate" between identifying significant aspects of the problem while proposing provisional ideas for a solution. Allowing them to "illuminate obscurity on both sides and reduce misfit between them" (Archer, 1979, p. 17).

This ability to oscillate between problem and solution that is anchored in a solution finding is best observed and described in the comparison of problem-solving approaches between natural science and design. One of the main differences between the two different approaches

lies in that scientists set out to study the problem with the intent of discovering the rule that governs the behavior and in which a single optimal solution can be found (Cross, 1982; D. H. Jonassen, 2000; Lawson, 2006). On the other hand, designers seek to understand problems with the help of a primary generator or temporary solution that acts as limits of what the problem space might be and what could be a possible solution (Cross, 1982; Darke, 1979). Later, through an iterative process of testing and evaluating these temporary solutions, designers redefine their understanding of the problem to propose new, better-fitting solutions (Cross, 1982; Darke, 1979). A more abstract analysis of this behavior points to design as a constructivist, pattern-assembling discipline and designers as actors who construct patterns by combining and rearranging elements that make a new solution that reduces misfits between the current problem state and a new desired one.

In light of this definition of how design and designers differ from practitioners from other disciplines, it is also important to note two of the most discussed paradigms of designerly ways of acting. On the one hand, the aforementioned Simon (1969 & 1973) describes design action as rational problem-solving rooted in his studies of decision-making and rationality (Dorst & Dijkhuis, 1995; Hatchuel, 2001). Simon regarded design as a creative endeavor in which practitioners sought to create solutions to ill-structured or ill-defined problems and in which an important part of the process is to provide structure to the problem itself (Hatchuel, 2001; Simon, 1973, 1996). He also equated the process to a search through a maze in which a satisfactory solution is found by selectively searching through a vast number of possible solutions (Simon, 1969, 1996). In which the solution seeker limits the search area by providing a definition or structure to the maze by dividing or simplifying it into smaller manageable problems (Simon, 1973).

On the other hand, and juxtaposed to Simon's view of the design practice, we find the reflection-in-action paradigm introduced by Schön (1983) and reiterated by multiple design researchers (You & Hands, 2019). This theory of designerly ways of acting sought to, among other things, contest Simon's view of design problems (Dorst, 1997; Dorst & Dijkhuis, 1995; You & Hands, 2019). One of the main criticisms of Simon's view rests on the description of design problems as given or existing and being in need of exploration rather than being constructed or framed in the midst of design practice through an act of reflection (You & Hands, 2019). This act of reflecting is characterized by the ability and willingness to continuously construct the problem to be solved by 're-framing' it, creating new questions and new desired outcomes during the design process (Dorst, 1997; Dorst & Dijkhuis, 1995; Schön, 1983; You & Hands, 2019).

Furthermore, Schön and his perspective on design ways of knowing and doing in reflection-in-action widely emphasize the importance of reflective thinking and learning in the design process. Reflection-in-action refers to the ability to think critically and make adjustments while actively engaged in the design process (Schön, 1983; Schwartz, 1987). This involves reflecting on the actions and decisions taken during the design process, as well as analyzing and evaluating their proposed solutions in light of the frames assigned to the problem (Dorst & Dijkhuis, 1995; Schön, 1983; Schwartz, 1987). Said reflection can take different forms, including reflection on action and reflection in action. Reflection on action involves looking back on completed design solutions and analyzing the outcomes and lessons learned. Reflection in action, on the other hand, involves thinking and making adjustments in real time during the design process (Lousberg et al., 2020).

2.2.3. Design Thinking Problem-based learning.

As design thinking has gained popularity in business settings, it has done so in education settings; scholars are increasingly examining its diverse applications and understanding both its strengths and limitations (Johansson-Sköldberg et al., 2013; Lor, 2017; Panke, 2019; Scheer et al., 2012). Educators are attracted to design thinking for its potential to facilitate engaging learning while at the same time helping to develop and attain desirable pedagogical goals. In Panke's (2019) literature review of design thinking within the educational setting, the author discusses the potential of design thinking in light of the effects that design thinking can have on education that goes beyond design thinking itself.

Design thinking education approaches can then play a significant role in providing students with multiple different types of learning, such as user-centeredness, iteration and experimentation, creativity and innovation, and problem framing (McLaughlin et al., 2019; Micheli et al., 2019; Rauth et al., 2010) that are interconnected to design practice principles. Furthermore, citing Melles & Co. (2015), Panke stresses the role of design thinking as 'complementary to existing pedagogies,' which helps to develop skills non-design exclusive to design practice such as interdisciplinary collaboration, empathy, cognitive bias reduction, tolerance of ambiguity and failure among others (Panke, 2019).

The Hasso Plattner Institute of Design at Stanford University, also known as the d.school, has been a pioneer in promoting design thinking concepts in higher education (Camacho, 2016) by incorporating and integrating principles from Problem-Based learning and Challenge-Based learning with aspects of human-centered

empathic design principles into engineering education (Dym et al., 2005). Furthermore, the incorporation of these principles in a setting of global collaborative projects has also helped to foster transversal skills through the multidisciplinary and collaborative environment of the projects (Dym et al., 2005; Jalali et al., 2022).

By introducing design thinking into engineering education, educators aim to provide students with a distinctively different approach to solving problems that differ significantly from the traditional technical or analytical approach typically used in engineering. The main differences between the two different approaches lie in that the traditional engineering approach sets out to solve problems in a scientific way in which problems are studied with the intent of discovering the rule that governs the behavior and in which a single optimal solution can be found (Cross, 1982; D. H. Jonassen, 2000; Lawson, 2006). On the other hand, designers seek to understand problems with the help of a primary generator or temporary solution that acts as limits of what the problem space might be and what could be a possible solution (Cross, 1982; Darke, 1979).

However, the use of design thinking in industry and education has also had its share of criticism and emphasis on its limitations. Some of the criticisms share principles similar to those of the PBL, on which the design thinking model is based. Mainly based on the quality of each individual program that spans from instructors and curriculum variability (Prince & Felder, 2006). This is also deepened by the critics of Design Thinking itself based on its simplification and dilution as a problem-solving methodology (IDEO, 2018b; Johansson-Sköldberg et al., 2013).

2.3. Objects of design

So far in this chapter, I have aimed to highlight the importance of the development of design knowledge and design expertise through experiential learning in design education and what it can contribute to the learning development of students. This has served to provide a theoretical foundation and context understanding to the different studies I have conducted, which relate both to why and how the objects we use to aid in the development of a designerly way of thinking and doing in multidisciplinary engineering education. In which has meant the examination of the different objects of design that allow students to develop experiential knowledge and develop a design ability to frame and solve problems. However, these same objects also help to bridge barriers of understanding and communication between students, educators, and other stakeholders that may arise from a multidisciplinary environment.

The following theoretical context that will be presented revolves around the different objects of design practice that I have studied. These objects in the educational setting of designerly thinking education serve as tools for problem exploration and sensemaking in ill-defined problems, as well as tools for educators to reinforce educational objectives such as team dynamics and collaboration in multidisciplinary contexts. They also serve the purpose of a boundary object that facilitates the interaction and alignment of the teammates around different stages of the problem-solving process.

In the case of this thesis, this examination of the boundary object of design that we utilize means to examine the way that problem is formulated in design briefs plays a role and has an impact on

how problems can be addressed and which design strategies can be utilized. It also means to examine further uses of prototypes in design education that allow for the exploration of problems and their implications to possible solutions. The use of ideation aids and tools that help to incorporate elements of randomness to both generate novel ideas and help to reframe problems under different unseen perspectives. Finally, to explore the tools, methods, and processes that are used to create teams and how they can be used to develop trust within teams to foster better collaboration and communication among individuals of the team.

2.3.1. Boundary Objects

The term boundary object was first introduced by Star and Griesemer (1989) in the field of Zoology to refer to objects that facilitated and bridged different perspectives to aid communication between different social groups or disciplines (Carlile, 2002; Stoytcheva, 2013). In the context of design engineering, boundary objects could be interpreted as artifacts, tools, or methods that facilitate communication, collaboration, and knowledge sharing between different stakeholders involved in the design problem-solving process (Bergman et al., 2007; Hussmo et al., 2022; Pierre Johnson et al., 2017; Rhinow et al., 2012). These objects act as a common reference point that helps bridge the gaps and differences in understanding and perspectives among various stakeholders.

Boundary objects are particularly useful in situations where knowledge is complex and multifaceted and where different stakeholders have diverse perspectives and expertise (Carlile, 2002). They provide a shared platform for interaction, allowing individuals from different backgrounds to find common ground and work

towards a common goal by serving as a common reference point to align multiple perspectives and facilitate the exchange of knowledge and information (Carlile, 2002).

Boundary objects in design engineering can take various forms, such as prototypes, design specifications (such as design briefs), or design methods, among others (Wenger, 2000). These objects serve as tangible representations of ideas, concepts, or requirements, allowing different stakeholders to interact, exchange information, and align their understanding (Carlile, 2002; Wenger, 2000). The use of boundary objects in design engineering is particularly valuable when dealing with complex and multidisciplinary projects, where different teams or individuals with diverse expertise and perspectives need to collaborate effectively (Hussmo et al., 2022).

2.3.2. Boundaries

Boundaries are understood as "real or imagined lines that mark the edge or limit of something" (Cambridge Dictionary, 2024). This definition provides an interesting notion that boundaries are not only real limits or lines which can be easily marked or referenced in the physical plain, but they can also be imagined or rather imaginary intangible limits. In the case of social interactions and communication, boundaries and can be understood as limits that separate different groups. Wenger (2000) presents us with a clear exemplification of said boundaries.

"Sit for lunch by a group of high-energy particle physicists and you know about boundary, not because they intend to exclude you but because you cannot figure out what they are talking about. Shared practice by its very nature creates boundaries." (Wenger, 2000, p. 232).

Furthermore, boundaries are not only important from the perspective of conceptual divisions or interruptions that "give rise to discontinuities in interaction and action" (Akkerman & Bakker, 2011, p. 139). They can represent points of intersection where different groups that possess different knowledge can come together and try to find a shared and common meaning (Carlile, 2002; Star & Griesemer, 1989; Wenger, 2000). And it's in this intersection where boundary objects, as previously presented, serve as the bridges to help mediate, translate, and facilitate a common understanding between groups (Akkerman & Bakker, 2011; Carlile, 2002; Hussmo et al., 2022; Star & Griesemer, 1989; Stoytcheva, 2013; Wenger, 2000).

2.3.3. Boundaries in Education

When discussing formalized higher education, we usually refer to education that happens at the level of learning that occurs at universities, colleges, and other institutions that award academic degrees. It is typically divided into separate and distinct fields, each focusing on specific areas of study and research. Each field has its own unique body of knowledge, methodologies, and theories that are taught and researched within academic institutions. This division allows for specialization and expertise in particular areas of study, contributing to the advancement of knowledge and the development of skilled professionals (Schofer & Meyer, 2005).

This separation into distinct fields in itself has been of vital importance for human development as it contributes to the advancement of knowledge and the development of skilled professionals in various industries. Which at the same time leads to societal progress and economic development (Schofer & Meyer, 2005). However, the more specialized we become and the more compartmentalized knowledge gets, the bigger the boundaries between professionals from separate fields (Akkerman & Bakker, 2011).

In the context of interdisciplinary collaboration and knowledge sharing, boundaries play a crucial role in defining the scope and limitations of shared understanding and communication (Akkerman & Bakker, 2011; Carlile, 2002; Wenger, 2000).

Beyond a perceived increased interest in interdisciplinary and multidisciplinary collaboration among fields during the 21st century, as evidenced by the growing calls from academic institutions, federal agencies, governmental and private funds, as well as from academics themselves (Jacobs & Frickel, 2009). Multidisciplinary collaboration is not a particularly new idea; rather, knowledge has always been produced in the context of collaborative academic spaces, which has also led to the creation of new disciplines based on a combination of existing ones (Barry et al., 2008). However, this growing call for more multidisciplinary development seems to come not necessarily from an academic need but rather from a governmental and organizational-driven push for innovation and economic growth (Barry et al., 2008).

As such, the demand for professionals who can navigate and integrate knowledge from diverse fields but, more importantly, who are capable of collaborating with professionals from other fields has led to a growing emphasis on interdisciplinary and multidisciplinary education. Overall, the shift towards multidisciplinary education is a response to the demand for professionals who can navigate complex challenges in which a single type of expertise will not be sufficient to find a solution (Klaassen, 2018). Rather, it involves drawing on diverse bodies of knowledge and expertise from multiple participants and professionals (Klaassen, 2018).

In these kinds of problem-solving environments, the use of boundary objects in educational development plays a crucial role as they facilitate the interaction and collaboration between different communities or

groups within the educational context (Akkerman & Bakker, 2011; Wenger, 2000). This with the intention of creating educational tools that foster the development skills such as collaboration, communication, negotiation, among other (Wenger, 2000).

These objects can be physical or conceptual and are used as a means to bridge the gap between different perspectives, knowledge, and practices. In design engineering, they can take the form of models, drawings, prototypes, design documentations, methodological tools, etc. (Bergman et al., 2007; Hussmo et al., 2022; Pierre Johnson et al., 2017; Rhinow et al., 2012). They can aid not only in collaboration between students from different fields but also in student-to-teacher interactions and teacher-to-teacher interactions (Akkerman & Bakker, 2011). Furthermore, in the case of project-based learning with industrial partners, it can help to bridge communication between corporate stakeholders and academic stakeholders to align the language and practices of the educational process and foster a shared understanding and common goals.

2.3.4. Types of boundary objects

Since Star & Griesemer first introduced the concept of boundary objects in 1989, multiple academics have also studied and attempted to characterize and classify boundary objects into different groups (Carlile, 2002; Star & Griesemer, 1989; Wenger, 2000). However, for this thesis, I will mainly focus on the classification provided by Wenger (2000), given that his characterization of boundaries and boundary objects is rooted in learning communities. Wenger presents us with a characterization of boundary objects that are based on social learning systems and the concept of communities of practice. Where the cognitive process of learning occurs through collaborative and interactive learning within social contexts (Wenger, 1998 & 2000)

and in which learning is "defined [as an] interplay between social competence and personal experience" (Wenger, 2000, p. 227)

In his essays, Wenger (2000) presents us with a characterization of the effects that can take place where interactions between groups from different 'communities' occur. These occurrences are referred to as boundary processes, which Wenger characterizes based on the types of bridges that help to cross said boundaries. First, Wenger (2000) presents us with the following dimension for the effects that occur in boundary process:

- Coordination to enable collaborative action,
- Transparency, which provides access to meaning or understanding,
- Negotiability, which provides space for compromise and agreement.

Secondly, Wenger (2000) describes three types of bridges to cross those boundaries: people, objects, and interaction. I will focus mainly on the objects for the purpose of this thesis. To do so, we will once again look at Wenger (2000) and the different types of boundary objects he presents as elements that support the interaction in boundary processes. In describing boundary objects, Wenger provides us with extra dimensions of the objects based on the form that they can take:

- Artifacts like tools, documents, prototypes,
- Discourses that provide a common language,
- Processes that allow coordinated action

Finally, Wenger (1998) and expanded by Filippi et al. (2023) also provide us with four further dimensions which give an extra understanding of the objects in terms of their intrinsic value and the abilities of the object:

- Abstraction allows the object to be understood from multiple perspectives and serves to facilitate dialogue between groups.
- Accommodation allows for the object to be used in multiple activities.
- Modularity allows the object, or pieces of the object, to be used in different situations.
- Standardization allows the object to be directly interpretable and used.

These three different levels of categorization or characterization of boundary objects, and the effects that they have in bridging boundaries, provide a platform for later examination of the different objects created in the studies of this thesis using both the type of object, their intended bridging function, and the intended effect that it has in the educational practice.

Methodology

03

Chapter 3

3. Methodology

Before delving into this methodology chapter and discussing the general approach that I used in my research and in this thesis and the specific methods used in the different studies, I would like to first give a general introduction to my particular process. For those familiar with my licentiate thesis, which I hope to be several of the readers since it is appended in this Ph.D. thesis, you will find this introduction somewhat recognizable. As stated in it, the intention of this introduction is to provide a look at my process and reasoning.

A big part of my professional life and my vocational call has gravitated around teaching and education. Differently from some of my peers, I came into academia from a love of both design practice and the teaching of design rather than a purely research academic pursuit. During the last decade, I have been teaching design and product development in projects that are part of an international network of universities in a framework that is a multidisciplinary, project-based, and international learning environment. My involvement in these specific projects, as well as in different design-related courses, have been the prime driver and motivation for my research. Furthermore, within these projects, I have been involved with both of my supervisors in the further development of the methodological foundation and

the methods that we use while teaching the courses. This means my research and the different publications presented here are intricately connected to this very practical setting of an ongoing educational model.

3.1. Educational context

In 2016, Linköping University initiated the Design Thinking and Multidisciplinary Projects (DaMD) course, inspired by Stanford University's ME310 Global Engineering Design Innovation course. DaMD is part of an international collaboration with Stanford's ME310 and the SUGAR global network of universities. Projects are undertaken in multidisciplinary groups involving engineering, design, and strategy master programs.

The projects and the course are part of an international collaboration with both Stanford's ME310 and the SUGAR global network. These projects involve corporate sponsors and universities addressing real challenges through human-centered design. Typically, the setup of each project involves the combination of two universities, each of which provides a group of students from multiple disciplines to form a combined design team to work on the project. Each project is commissioned and funded by a corporate sponsor, which also provides a real challenge or problem that the team is said to work on during the project.

The DaMD projects are framed as an Innovative Problem-based course where students engage in collaborative and active, hands-on, participatory learning, working on ill-defined and open-ended problems. Furthermore, students acquire knowledge, and special emphasis is placed on problem framing and exploration as key aspects of the learning objectives. Each project addresses unique

challenges, fostering exploration of the problem initially presented by the organization as well as an exploration of user needs. While this diversity offers advantages, it also means that each team faces a very unique challenge.

In 2021, we started transforming our course as part of a Vinnova-funded project to develop a new educational model project in collaboration with SAAB AB. The project intended to further develop the DaMD course into an Open Education Platform (OEP), aligning with national efforts by 'Teknikföretagen' and the Swedish Ministry of Education. The OEP provides work-integrated learning for students and lifelong learning opportunities for practitioners in the industry. In the new set-up, regular students engaged in the same DaMD course, while practitioners from industry had a flexible commitment that allowed them to participate in some or all aspects of the course, including coaching, workshops, and teaching units for competence development. The program fosters collaboration between students and practitioners, promoting relevant education and ongoing professional growth for both students and practitioners (Wever et al., 2023).

3.2. Philosophical and epistemological stance

Based on the notion that the studies presented in this thesis are situated in a very practical environment of ongoing teaching practices within university/industry collaboration projects, this has important implications for the methodological approach. My general approach to research is very much rooted in my own education and experience in design and education. Placing a high importance on problem-solving and the practical implications (and applications) that my research has on my own teaching. With this in mind, my philosophical stance is rooted in pragmatism, which emphasizes the practical implications

and the broader value of the research (Kaushik & Walsh, 2019; Maxcy, 2003; Meissner et al., 2011; Rorty, 2000). However, given that pragmatism is not necessarily one single clear philosophical stance but rather encompasses different schools of thought (Dalsgaard, 2014), it is important to mention that my view aligns more closely with Deweyan Pragmatism.

Deweyan pragmatism is a philosophical framework characterized by its emphasis on instrumentalism, experimentalism, situated practice, and reflective inquiry (Dalsgaard, 2014). This translates to a focus on the real-world implications and applications of theoretical frameworks and methodologies and a reflective and adaptive approach to the research process (Säfsten & Gustavsson, 2020). More importantly, from Dalsgaard (2014), we can draw parallels between the core principles of design and Deweyan Pragmatism, in which both promote a view that reiterates that practical application and testing are of utmost importance for concepts to demonstrate their worth. Both place a strong emphasis on active participation and highlight that knowledge is shaped through engagement with the situation. And hypotheses and concepts evolve through experimentation and reflection in practice.

In terms of its ontological and epistemological stance, Deweyan pragmatism presents a view of reality that emphasizes the dynamic, constantly changing, and experiential nature of existence, which rejects the notion of a fixed reality (Dillon et al., 2000; Johnson & Onwuegbuzie, 2004; Säfsten & Gustavsson, 2020). Instead, it proposes that reality is continuously shaped and reconstructed through human experience and inquiry (Dalsgaard, 2014; Martela, 2015; Säfsten & Gustavsson, 2020). Finally, in reference to methods and approach to research, Deweyan pragmatism's approach is characterized by an

emphasis on the practical consequences of knowledge, the value of knowledge in addressing real-world challenges, and the dynamic nature of knowledge production (Dalsgaard, 2014).

Finally, regarding my stance on methodological aspects of logic and scientific ideals made during the development of my studies, which relate to the relationship between theory, empirical data, and the conclusion that can then be drawn (Säfsten & Gustavsson, 2020) can be characterized as design abduction. This once again relates to the pragmatic view in which "a hypothesis is confirmed by means of its practical consequences [and] its usefulness" (Säfsten & Gustavsson, 2020, p. 55). This pattern of logic reasoning, and in my case researching, has been described and presented by Dorst (2019) to describe a designerly way of problem-solving. Dorst (2019) proposes that there are four basic patterns for reasoning that can be expressed in formal logic. They are based on how the elements (WHAT) and the connections between them (HOW) and the outcome create an effect (OUTCOME). From this, the four basic patterns for reasoning, deduction, induction, abduction, and design are abductive, which can be understood and expressed through the unknowns on the following equation.

WHAT + HOW leads to OUTCOME

In design abduction, at the beginning, we have limited knowledge of both the elements (WHAT) and connections (HOW) that allow us to reach a resolution. However, we do have an idea of the desired value (OUTCOME). This means that the process of creating or understanding an effect is by first proposing a temporary frame for a new how in order to be able to propose a new what. While iteratively, both unknowns are reframed, and new hows and whats are created until a correct 'fit' between the two can be reached (Dorst, 2019)

???? + ??? leads to OUTCOME

This, in terms of my research, means that in most cases, the starting point was not from an existing WHAT or HOW but rather to a desired perceived outcome or value that we wanted to see represented in the projects, in the students, or in the learning goals.

3.3. General research approach

As previously mentioned, the research that I have conducted during my doctoral studies has been conducted within central platform of both the Design Thinking and Multidisciplinary Projects course and the Open Educational Platform research project in which I have participated in developing. This means my research and the different publications here presented are intricately connected our ongoing work within this educational model.

In terms of my general research approach and how those methods are aligned, my research can be situated within two very commonly used research approaches in design research. Firstly, my research can be situated in Action-Research, which is a systematic research approach rooted in problem-solving and improvement within a specific context and which involves researchers and practitioners in the context of the problem (Säfsten & Gustavsson, 2020). Secondly, in research through design, a research methodology that advocates for the use of the act of designing, the creation of tangible artifacts, and reflection in the generation of knowledge (Gaver, 2012).

Moreover, and core to Pragmatism, Action-research, Research Through Design, and general design practice, they all aim to deal with

and solve wicked problems (Buchanan, 1992) where the designer strives to find a satisficing solution (Simon, 1955) for the current context and state of the problem rather than a discovery of absolute optimal solution or undeniable truths (Gaver, 2012; Zimmerman et al., 2010). This, in principle for design research (and my research), can be argued that in itself is a stance on how design science and research are produced and evaluated in somewhat conflict with Popper's description of science based on the principles of falsifiability and validity (Gaver, 2012; Zimmerman et al., 2010)

3.3.1. Action Research

Action research, first introduced by Kurt Lewin (1944, 1946), integrates research and action to bring about practical solutions to issues encountered in a specific context. It is characterized by its participatory nature, involving the collaboration of researchers and practitioners to identify problems, develop interventions, and evaluate outcomes (Holter & Schwartz-Barcott, 1993). This approach emphasizes the importance of bridging the gap between theory and practice and in which the researcher reflects on their actions to better justify them in the context in which the action is carried out (Holter & Schwartz-Barcott, 1993; Säfsten & Gustavsson, 2020).

Overall, action research is an iterative process that allows practitioners to improve practice and contribute to the development of practical knowledge (Carr & Kemmis, 2003). in which practitioners and researchers engage in a multi-cyclical process of planning, action, observation, and reflection (Figure 8) to solve practical problems (Altrichter et al., 2002; Carr & Kemmis, 2003; Holter & Schwartz-Barcott, 1993; Säfsten & Gustavsson, 2020).

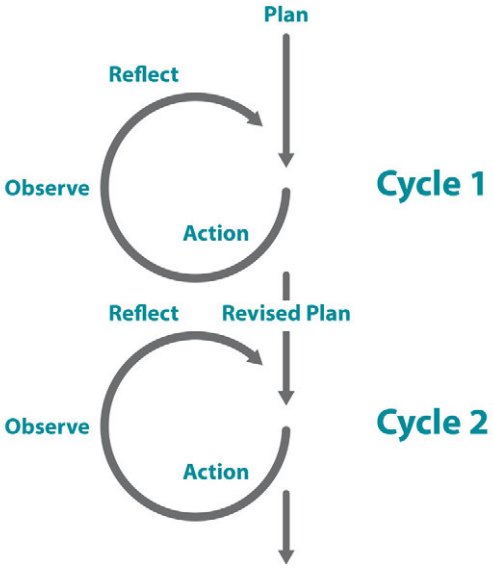


Figure 8 Action research process (based on Kemmis & McTaggart by Säfsten & Gustavsson, 2020)

3.3.2.Research Through Design

Research through design (RtD) is a concept attributed to Christopher Frayling's paper "Research in Art and Design" (Frayling, 1993). Frayling's paper introduced the idea of three modes of artistic research: research in art and design, research through art and design, and research for art and design. Frayling's (1993) work emphasizes the significance of practice-based research in art and design, highlighting that the knowledge and skills of a practicing artist or designer form a central part of the research process and in which knowledge is embodied in the artifacts themselves.

However, Archer (1995) and Findeli (1998) further develop the concept to provide more grounding and clarification based on design

research and principles, which themselves connect to the principles of engagement, action, and transformation of problems within a context (Jonas, 2015). Furthermore, it emphasizes the creation of knowledge through reflective design practice and the making of physical artifacts, which more closely resemble the view of design action presented by Schön (1983) as a reflection in action. In a further development of these core ideas and relying on the seminal aspects of design research by Archer, Owen, and Findeli, Jonas (2015) challenges and expands some of the original ideas of research through design presented by Frayling. Firstly, Jonas (2015) argues that the main principle of RtD is not the creation of artifacts but the designerly process that leads to the generation of knowledge, which utilizes objects not as carriers of knowledge but as facilitators. Secondly, an important distinction between design itself and researching through design:

> "Design uses and applies analytical knowledge for the purpose of designing artifacts or finding answers to design questions, whereas RtD uses designerly, project-based process knowledge for the purpose of finding answers to research questions." (Jonas, 2015, p. 34).

3.3.3. Why Action Research & Research Through Design

Drawing from Deweyan pragmatism, the justification for both action research and research through design becomes apparent. Deweyan pragmatism highlights the significance of practical experience and problem-solving in the pursuit of knowledge. Aligning with this perspective, action research involves a systematic inquiry by practitioners to address real-world problems and enhance practices within their specific context (Dalsgaard, 2014; Martela, 2015; Säfsten & Gustavsson, 2020). Likewise, research through design aligns with Deweyan pragmatism by underscoring the importance

of experimentation and the integration of theory and practice in the creation of knowledge (Dalsgaard, 2014). Consequently, embracing action research and research through design for my own pursuit of design education can help foster a practical and experiential approach to knowledge acquisition and application.

In the development of my research project during my PhD studies both Action-Research and Research through Design were used in the process of all my papers to some extent not as a rigorous adherence to the methods but rather as my own designerly thinking and doing behavior. This, in practical application, means several things:

- Firstly, the main purpose of my studies was, to a certain extent, to address 'problems,' dissatisfactions, or areas of opportunity in the frames, methods, or tools used in the process of designerly education.
- Secondly, my studies followed an iterative process in which I oscillated between proposing frames, methods, and artifacts to then bring them to action in testing with students or other faculty members. Then, bring those findings and observations to reflect on them and either propose new frames, methods, and artifacts or reach a satisfactory answer to the questions posed.
- Thirdly, all my studies were conducted in the context of the projects, with collaboration, observation, or participation from practitioners, may they be students going through the educational process or faculty members from Linköping University and other universities that utilize the same PBL approach to education.
- Finally, this meant that the main purpose of my research was to generate useful and usable knowledge by means of using boundary objects in the attainment of the educational goals of the projects.

3.4. Research methods in the different studies.

3.4.1. Typological study

It is important to note that in my licentiate thesis, the main objective was to develop a typology of design briefs and not a taxonomy of briefs. A typology is a classification system that categorizes entities based on shared characteristics, allowing for the identification of patterns and relationships within a particular domain (Doty & Glick, 1994). Typologies are often used in various fields to organize and understand complex phenomena (Dingemanse, 2019; Doty & Glick, 1994; Marradi, 1990). On the other hand, a taxonomy is a hierarchical classification system that arranges entities into broader and more specific categories based on their shared characteristics (Bailey, 1994; Doty & Glick, 1994; Marradi, 1990). While both typologies and taxonomies are used for classification, typologies tend to focus on capturing the central characteristics of a phenomenon, while taxonomies are more extensive and often involve hierarchies among categories (Bailey, 1994; Doty & Glick, 1994; Marradi, 1990). In essence, typologies aim to identify commonalities and differences among entities, while taxonomies aim to create a structure for organizing entities based on their attributes (Doty & Glick, 1994).

3.4.1.1. Licentiate thesis

The Problem of Formulating Design Problems: A Typology of Design Briefs

The process that followed in the creation of a typology of design briefs was an iterative process that involved a codification of the data (design briefs) to find patterns in the different information that the briefs contained to form a cluster that led to the typology (figure 9).

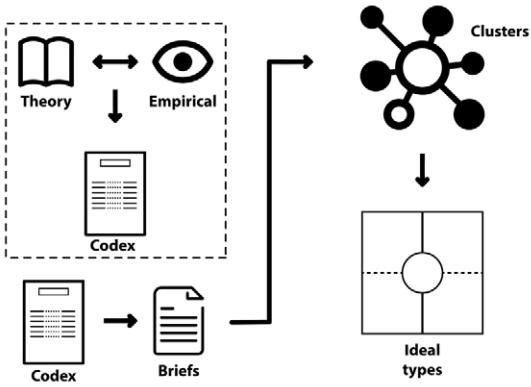


Figure 9. Typology creation process

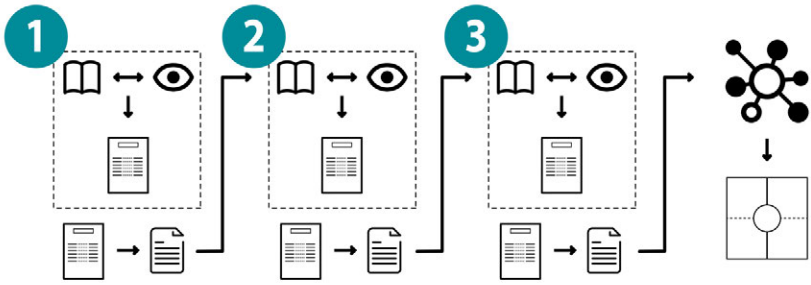


Figure 10. Typology creation process based on number of iterations.

However, this was done through three iterations of the codification system with the use of brief characteristics from both theoretical aspects of problem formulations as well as practical aspects of the document itself (figure 10).

The process of refining the codex followed an iterative prototyping process in which new characteristics were introduced or removed as more briefs were codified. After all briefs were codified the next step in the process was to use the findings along with the theoretical aspects of problem structure presented in literature to define generalizable constructs that allowed for the construction of ideal

types that represented "complex constructs that can be used to represent integrated configurations of unidimensional constructs" (Doty & Glick, 1994, p. 233).

It would be easy to linearly describe the process in which the methodology was crafted, and then the data set was collected. However, the reality of the process, is that the type of data (namely the design briefs) to which I had access drove the decisions made in terms of the methodology and how it was used in the development of the typology.

3.4.2. Interviews and Thematic Analysis Study

Semi-structured interviews are a common method used in qualitative research to gather in-depth and rich data from participants (Cohen et al., 2017). Unlike structured interviews, semi-structured interviews use open-ended questions and a non-rigid interview structure that allows more flexibility and exploration of new topics that may come up during the interview (Bogdan & Biklen, 1997; Portigal, 2013; Säfsten & Gustavsson, 2020). This method allows researchers to delve into participants' perspectives, experiences, and insights related to design processes, challenges, and outcomes. By conducting semi-structured interviews, paired with a data analysis method, such as thematic analysis, it is possible to gain a deeper understanding of the complexities and nuances of design processes and the behaviors of the practitioners.

Thematic analysis is a widely used qualitative research method in multiple fields beyond design. This method involves the identification, analysis, and discussion of patterns or themes in different types of qualitative datasets (Castleberry & Nolen, 2018; Peel, 2020; Säfsten & Gustavsson, 2020). The thematic analysis method can be applied

to various types of data, including focus groups and historical documents. However, the most common use of thematic analysis is related to the analysis of structured and semi-structured interviews. In the context of design research, thematic analysis can be used to identify recurring themes or patterns in participants' experiences, perceptions, and attitudes toward design processes, products, or interventions (Peel, 2020; Säfsten & Gustavsson, 2020). Thematic analysis of expert designers, for example, in the case of my research, themes can be used as a discussion tool that is meant to represent the voice of the participants as it intended to showcase the underlying behaviors while being contrasted with theoretical underpinnings of the design practice to better understand the behaviors and mechanism at play.

3.4.2.1. Paper I

Design team formation using self-assessment and observer-assessment techniques ; mapping practices in a global network of universities.

The study explores team formation in design and engineering project-based education by interviewing ten experienced teachers from seven different universities involved in project-based courses. The teachers had experience forming teams consisting of 15 to 100 students, creating 2 to 10 teams per university, depending on project availability. Semi-structured interviews averaging 60 minutes were conducted, focusing on the team formation process, criteria used, and team creation process.

A semi-structured interviewing protocol was employed, focusing on three main areas: the activities used in team formation, the criteria observed during these activities, and the process of creating teams based on the observed criteria. Furthermore, the creation and

development of the protocol itself were done in an iterative way in which the protocol was enriched based on the previous interviews when unforeseen areas of interest were discovered (figure 11).

The use of semi-structured interviews allowed for an exploration of the participants' experiences and perspectives, contributing to a comprehensive understanding of team formation methods within the context of teaching teams. The interviews were analyzed using qualitative analysis once again in an iterative process of codifying the findings of the interviews by exploration of practices and themes, where the codification was constantly modified based on findings

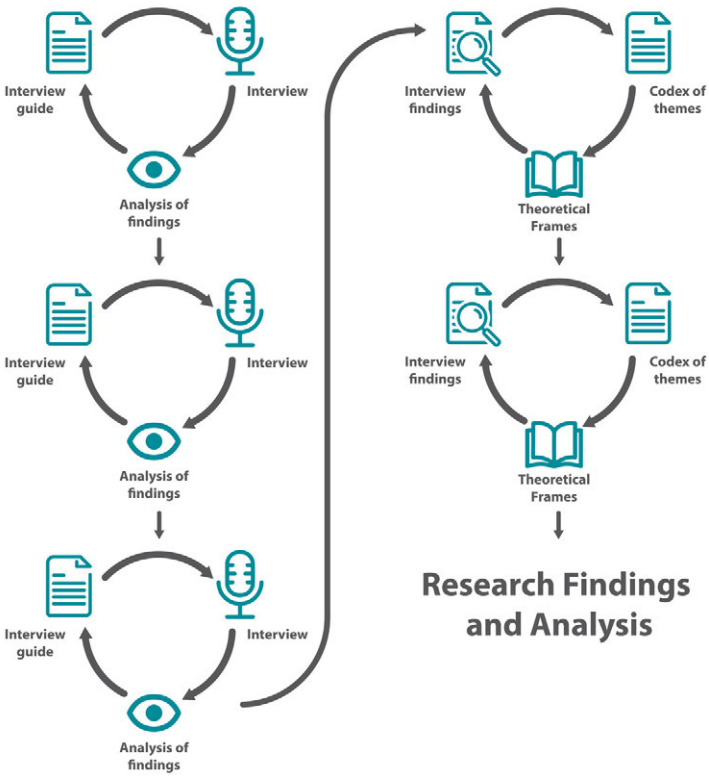


Figure 11. Paper I creation process with multiple iterations

of the different interviews and new theoretical frames that could be applied to the newly discovered themes (Figure 11). This led to the identification of themes such as practitioner's expertise, small-scale experimentation, and learning from experiences. These themes shed light on observer-assessment team formation processes and their differences from self-assessment team formation tools.

3.4.3. Research through Design

Research through design (RtD), as previously mentioned and explained, is a methodology that involves the integration of design practices and research activities to develop new knowledge. And which has been used as a method for design research, particularly in addressing wicked, ill-structured problems and developing design theory and methods (Zimmerman et al., 2010). However, it is important to clarify that RtD stands out from other methodological approaches, especially those of the natural sciences. Firstly, based on how research questions are conceived and secondly, because of the medium in which the questions are explored.

In regard to this first characteristic of RtD, Findeli (2010) helps us to identify a model and medium in which research through design can help to generate research in design through the use of design questions that can be constructed into research questions. He argues that research in design often starts from an observed dissatisfaction within a routine or mundane daily context, which the designers are adept at identifying, as it is within this realm of everyday complex experiences that design aims to find problems and develop solutions. This identification of design problems or design questions then leads to a rather important step of transformation of this contextual design question into a generalizable and fundamental human experience problem (Findeli, 2010). Consequently, these contextual design questions are then turned into research questions (figure 12).

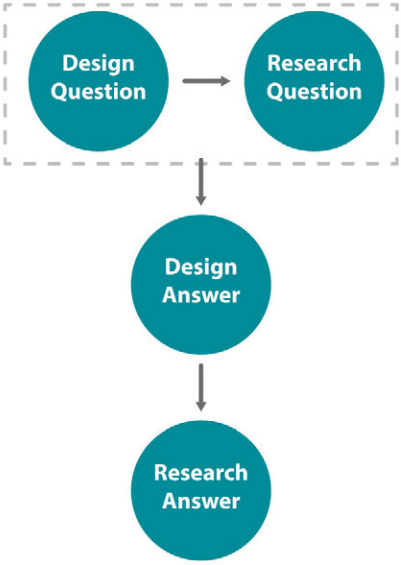


Figure 12. Research Through Design process (Based on Findeli, 2010)

Now, to address the second characteristic that differentiates RtD, we look at the way in which those questions are addressed. In order to do this, we might need to look to Jonas and the previously stated principle of RtD once again, which "uses designerly, project-based process knowledge for the purpose of finding answers to research questions." (Jonas, 2015, p. 34). This points to the act in which the answers, that produce answers to the research question, are created in the process of production of answers to design questions. However, as explained by Basballe & Halskov (2012), the RtD process follows a sequence of three steps: coupling, interweaving, and decoupling. In coupling, both the design project and research project go through a stage of development that serves both projects. During the interweaving stage, the development of both projects influences each other. Finally, and more importantly, a decoupling stage where both interests need to be separated, which for the process of the research project evaluation and inquiry become important to produce generalizable academic

knowledge. Furthermore, the overall process of RtD (Zimmerman et al., 2007) is underpinned by a constant reflection-in-action and reflection-on-action (Schön, 1983). A process of both a reflection after each action taken in the process and a reflection of the overall significance of what has been done. The complete process can be visualized in Figure 13

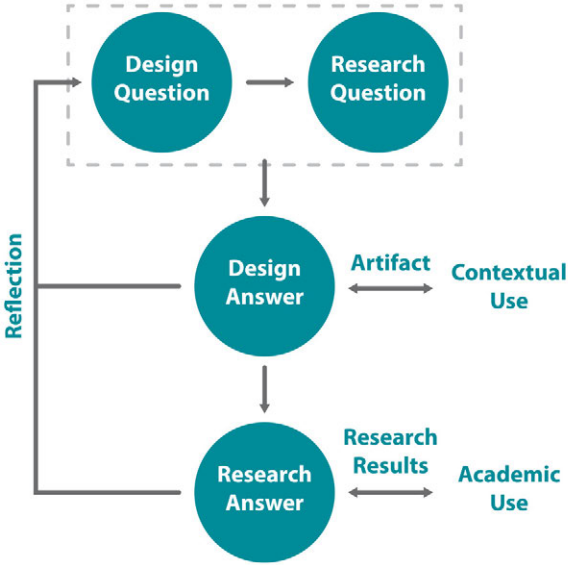


Figure 13. Research Through Design process expanded (Based on Findeli, 2010).

This process was then used in three of my papers, which are included here, not with the intent of using the artifacts and the design answers as undisputable evidence of a designerly behavior to replicate, but rather to use the situated practices in a specific context to then analyze them in light of the theoretical underpinnings and the underlying mechanism at play. The end goal of doing this was not to provide an optimal solution or artifact to use in ideation, prototyping, or trust building but rather to showcase how the artifacts have been used and to provide a theoretical understanding or foundation to practitioners that could shape their own practices.

3.4.3.1. Paper II

Into the woods; building trust in teams through scaffolded adventures

For Paper II, the initial design question came from a recognition or empirical observation that certain teams created with a balanced set of characteristics, both in terms of project-related and transversal skills, did not function well based on a lack of cohesion and trust in one another. In certain cases, it was evident that infighting and distrust were generated between students who came from different disciplines and master programs and had not established previous bonds before coming into the projects. This led to starting at a baseline of distrust, which, through negative interactions, was further accentuated. Based on these observations, the research question that followed was how to build a baseline of trust utilizing activities that promoted team bonding in the early stages of teamwork.

In order to find design answers and, consequently, research answers to the previous questions, we use an example of a scaffolded adventure-based trust-building exercise to observe a trust-building process to identify the underlying mechanism of how trust was built. The exercise is part of a master's program at Linköping University in which students were taken into an excursion day to a forest outside of Linköping. During the day, students participated in a number of activities, which, among others, included a scavenger hunt in underground caves and building and sailing a raft through a lake. Based on this, we looked at the excursion day process and the different individual exercises and activities included in order to describe and discuss the different trust-building mechanisms that each activity fomented.

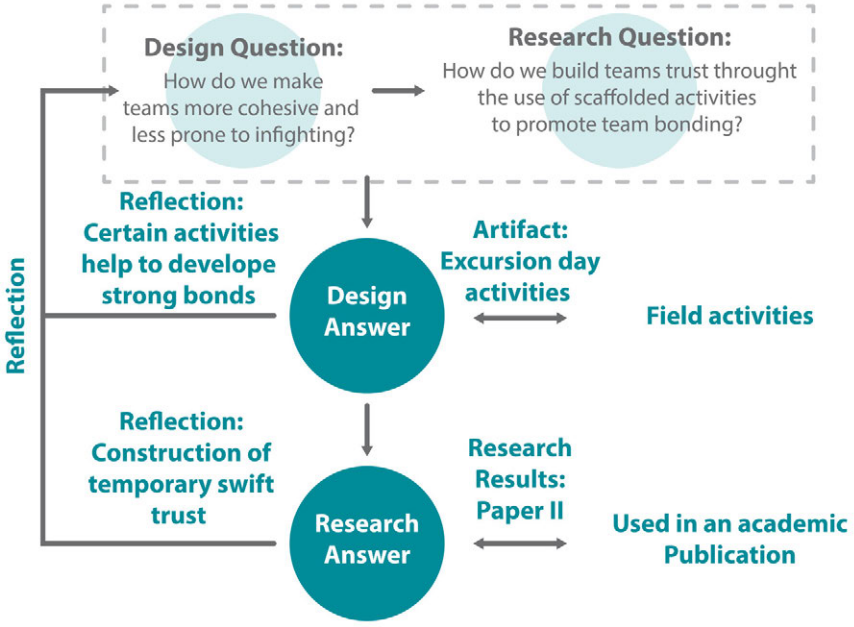


Figure 14. Research Through Design process in Paper II.

3.4.3.2. Paper III

This is How We Roll, a play on creative thought and the generation of novelty.

During the process of paper III, the initial design question or recognition of dissatisfaction with the ability of our students to find new ways of framing the problem of finding new uses for UAVs that went beyond the 'obvious' existing ones provided by either the design brief or the existing solution. This created a fixation on the initially proposed frames that the brief provided, which limited the student's ability to conceive new uses. From this point, the research question of how to both prevent fixation and facilitate an aimless and unguided

exploration that allowed for the creation of a new and unforeseen frame was created.

The process followed the creation of a design answer in the shape of a game ideation technique that introduced the element of randomness by using story cubes, which then were to conceptualize produced ideas into new frames. The game was created in an iterative process of testing with the students enrolled in the course, and new rules and procedures were developed based on the findings and discoveries. Finally, the last step of the process was the evaluation of the results of the game and their implication in the educational project that was based on theoretical concepts of fixation and the role of randomness in the process.

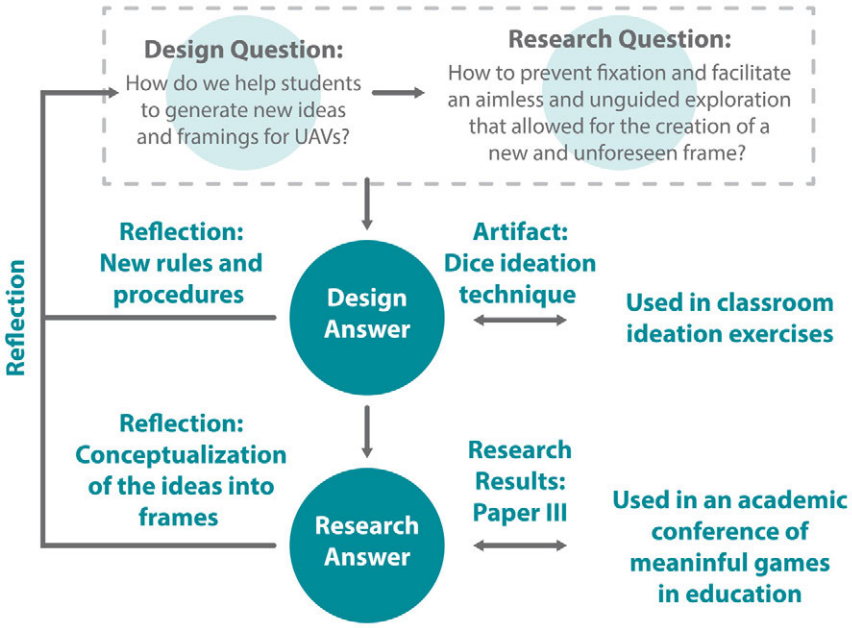


Figure 15. Research Through Design process in Paper III.

3.4.3.3. Paper IV

What is your (freaking) problem? Prototypes for problem exploration in the early stages of design

For the study conducted in paper IV, the initial design question was framed on a recognition of a point of struggle that students encounter in the early stages of the design process, which some navigated successfully while others didn't. As described in the paper, "One of the difficulties that students face during the first missions of the project is to produce prototypes that are not intended as fully fleshed solutions but rather as problem probing and exploration" (Ruiz & Wever, 2023, p. 706). However, some groups of students were able to create prototypes that helped them to fully reframe the problem based on discardable prototypes that were not the final solution but that had an impact in redirecting their efforts. Based on this observation, the research question that followed was "What does it mean to prototype with the (sole) purpose of problem exploration in the early stages of design?" (Ruiz & Wever, 2023, p. 703).

For this particular study, we as researchers did not create the artifacts that were used to produce solutions; rather, we examined the artifacts produced by the students during multiple years of projects in order to identify how these artifacts provided a solution to the research question. Based on this, the process of evaluation of the results to provide an answer to the research question was based on the theoretical concepts of prototyping in design and the identification of a gap in the literature that related to discovery-oriented prototypes. Based on this and utilizing the theoretical concept of reflection-in-action (Schön, 1983), we proposed an interpretation in which the point of reflection is not done on the frame proposed but rather on the overall context in which the frame exists and the implications it has on problem-oriented educational tasks.

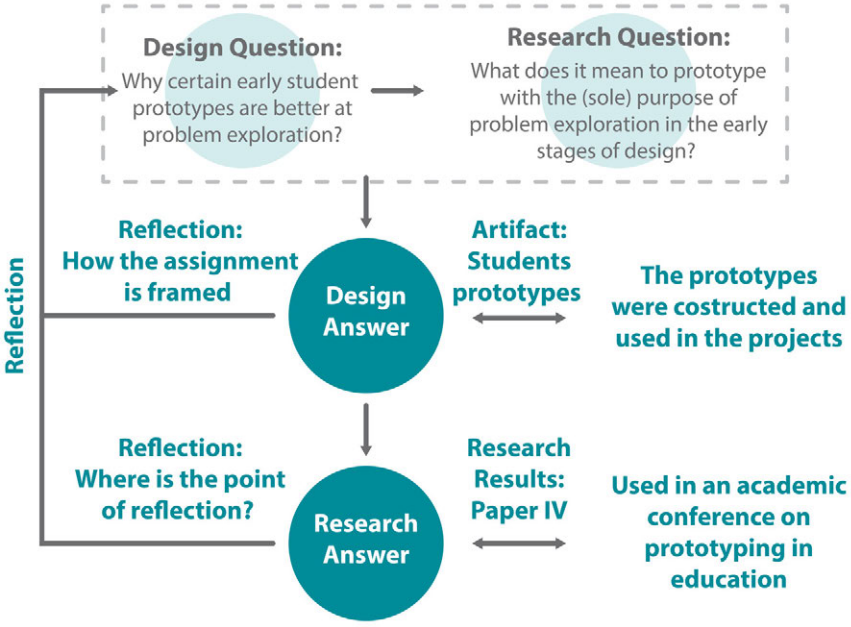


Figure 16. Research Through Design process in Paper IV.

Analysis and Contributions

04

Chapter 4

4. Analysis and contributions

In this analysis chapter, the different studies and an analysis and discussion of the findings will be presented, and the same order studies through the project process that are presented in the thesis introduction and in Figure 4. will be used. This will be done by first presenting a brief explanation, rationale, and a general overview of each study. Then, each of the studies will be discussed in light of the findings and outcomes, with an analysis of how each study relates to and contributes to the educational model as a problem-based learning structure. Subsequently, an analysis of the objects of each study as boundary objects that facilitate the multidisciplinary collaboration process. This analysis will be done by characterizing each object based on Wenger's (2000) characterization of the types of objects, the intended effect, and their intrinsic value.

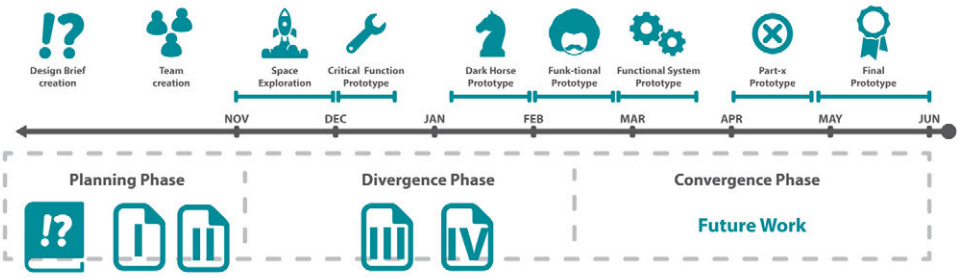


Figure 4. Studies of this thesis placed in timeline.

4.1. Project Planning Phase

One of the initial considerations and preparations that this type of educational approach needs to keep in mind are those that do not involve teaching per se but that carry significant implications. These initial considerations pertain to those of organization, administration, and bureaucratic matters. Among these are the projects that will be undertaken, the creation of design briefs that will be given to students to solve, and the selection, assessment creation, and building of student teams (figure 17). Given the multidisciplinarity nature of working with students from different programs and disciplines, there might be other special requirements of a multidisciplinary project-based educational model that need to be met and planned within regular higher education settings. Consequently, questions could be asked about how this type of education fits within the traditional model in future research, which falls outside of the scope of this thesis. Furthermore, other considerations such as project acquisition and external industrial partners' involvement need to be taken into account, and here, multiple other questions of academia-industry collaboration questions could be asked and developed for future research as well for future development and research.

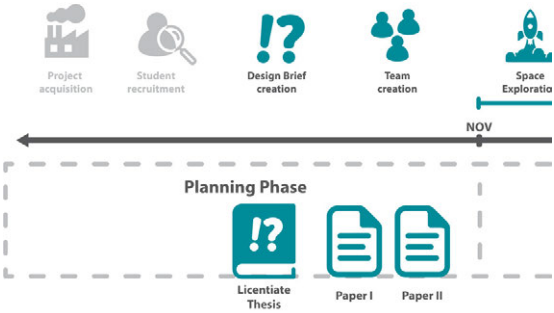


Figure 17. Project Planning Phase

4.1.1. Licentiate thesis

The Problem of Formulating Design Problems: A Typology of Design Briefs

The main intention of this paper was to examine the design briefs that served as the starting point of the projects. The desire to do so came from an empirical observation that certain teams of students struggled to start their projects as they struggled to determine who their main users were and what problems they faced. This struggle became more evident when teams from the network met for the Global Kickoff. In this instance, all student teams were asked to engage in the same assignment, yet they all started at very different levels of problem definition, which led to teams with more ill-defined problems struggling to start on their assignment. This empirical observation then led to defining the purpose of studying and understanding what constitutes a 'good' design brief. With the main intention of creating an empirically informed and theoretically underpinned typology of design briefs that characterized the different possible starting points and the kind of design problem-solving process that these starts could induce.

Design Briefs

At the beginning of any design project, there is usually the recognition or detection of a desire to change the current state of a problem into a new desired one (Chen, 2008). This recognition may manifest as the identification of an unmet need or problem, an identified opportunity, or dissatisfaction with current performance. This recognition in the pre-stages of a problem-solving process is usually translated into a document that aims to capture what is known about the current state

of the problem. This is usually done by the creation of a structured or semi-structured document that is then used to communicate the needs and requirements to be obtained (Ruiz, 2020). In the case of professional design practice, it is known as a Design Brief.

Beyond their simple use as a tool to either just communicate or as a point of validation between parties, the design brief carries significant importance for the design problem-solving process. As previously mentioned, design characterizes itself by dealing with Ill-structured problems that are poorly defined and in which information about what the problem might be is incomplete and conflictive (Buchanan, 1992; Cross, 1982; Rittel & Webber, 1973; Simon, 1973; Voss, 2005). Thus, it becomes clear that whatever limited information is known and how it is presented to the designers in the design brief takes further relevance as it will have a great impact on how the problem is approached and which strategy might be adapted to solve it.

From the perspective of design education and the model of designerly thinking education presented in this thesis, it is important to examine and understand problem formulation and the impact that it has in the attainment of the educational goals. In many cases where educational models similar to the one here discuss is utilized, there is also a differentiation in between the types of problems given to participants based on learning objectives.

Contribution & Analysis

Toward the educational model

As previously discussed in the literature review, there is a spectrum of educational approaches that utilize project-based learning and can have very different educational objectives based on the conditions

in which the project is situated. In projects where the intent relates more to interdisciplinarity and innovation, characteristics such as the type of problem provided and how it is framed will ultimately play a significant role. This further increases the relevance for educators to develop project briefs that are tailored to their level of problem definition, the type of learning objectives, the search process, and end results that are desirable to the course. Yet, this reiterates the distinction regarding the PBL model and its lack of a centralized definition and curriculum (Kolmos et al., 2009).

Based on this then the need to create briefs that are conducive to an exploratory process. That is both open enough to fit the need to allow the educational goals of learning to explore problem and solution but that also closed enough that can adder to the constrains of time and resources for exploration of problems and the production of feasible solutions.

The main intention of the study of my licentiate thesis was to explore and characterize the different dimensions of problem formulation and the effect they had on the overall search. With this in mind, I was able to produce a typology of four ideal types of problems, which presented distinct tradeoffs to the ensuing problem-solving process. Furthermore, according to the existing process of both our educational model of Linköping University and the process followed by both the ME310 course and the SUGAR Network, I provided a discussion and analysis of the types of problems and their structure was discussed in light of the desired educational goals and the structure of the course.

Design Briefing Tool as a Boundary Object

The subject of the design briefs as boundary objects can be seen from two perspectives. First from the perspective of the document

as a negotiation object between the organizations commissioning the projects and the teaching teams in charge of the projects. And secondly as a communication, coordination and negotiation object that helps in creating understanding and meaning between the multiple stakeholders (organization, students, and teachers).

Part of the contributions intended in my licentiate thesis was the production of a tool that aided the project stakeholders in thinking and discussing the starting point of the projects (figure 18). This tool contained elements that allowed for a discussion about what is known and unknown about the state of the problem and the expected process, deliverables, and timeline to follow. As a boundary object and based on Wenger's (1998 and 2000) work, we can characterize this tool as an artifact that provides the effect in the boundary process of transparency and negotiability, where both meaning and agreement can be developed. Furthermore, if we look at the tool based on its intrinsic value of having:

- Abstraction: to allow company representatives and teachers to have a dialogue around the problem to be provided to the students.
- Accommodation: to negotiate not only on this specific type of project but in other more traditional engineering education projects, as well as professional projects not connected to academia.
- Modularity: In which the different parts of the tool that defined different parts of the problem to be framed can be used separately from each other.
- Standardization: In which professionals from different fields, as well as educators from multiple fields can discuss the problems at hand.

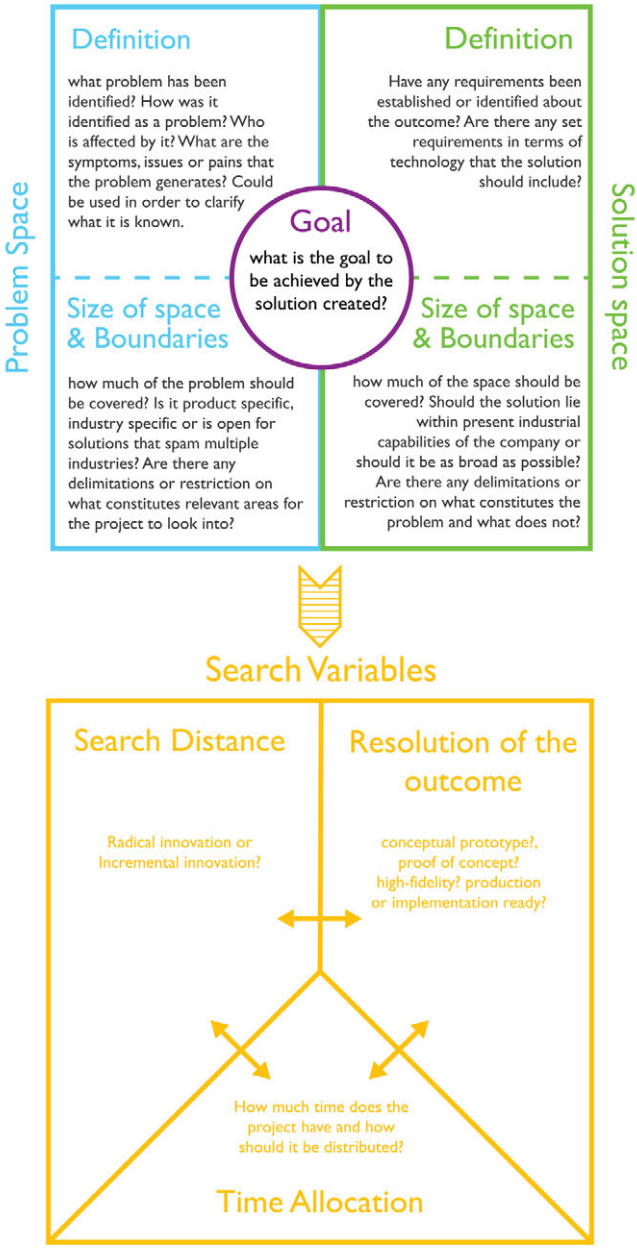


Figure 18. Design Brief creation tool from Licentiate thesis

Secondly, and perhaps more connected to the use of briefs as boundary objects that aid in the educational development of the students, the design briefs produced serve as multiple boundary-crossing objectives. The creation and use of design briefs that contain relevant and specific information about the problem can serve as a communication and negotiation object. Utilizing the document, students, teachers, and company representatives can discuss, understand, and come to agreements that are necessary for the development of the project. At the same time, the document must be developed and changed to add findings and new frames discovered during the process. Furthermore, if we look at the document based on its intrinsic value within the educational process, we can see the following:

- Abstraction: to allow all stakeholders to have a dialogue around the problem, its evolution, and new developments.
- Accommodation: The brief not only serves as a static document given at t=0 of the project process. Rather it can be used through the course of the project to capture new insights and knowledge but also as a sounding board that provides clarity of direction.
- Modularity: Different elements of the problem formulation such as knowledge of problem space and solution space can be used in different moments of the project process depending on the needs of the project. Moreover, the brief can be used as a starting point in research as well as reference point in decision making in accordance with the project needs.
- Standardization: The brief itself as a document can function as a problem framing tool, where interpretation and understanding from different perspective can provide new frames that one group of individuals can interpret while other don't which allows to further negotiation and development of the problem to be solved.

4.1.2. Student team assembly

As previously discussed during the theoretical framing of this thesis, one of the salient aspects of the educational model proposed and discussed here is that of creating a learning environment of multidisciplinary collaboration for students and other participants. Another important point that is being discussed within this thesis is that of boundaries and difficulties in communicating and collaborating in a setting where individuals from different contexts meet. In accordance with this, an important element of the type of educational projects that I have discussed pertains to teamwork and team collaboration that occurs during the project's process. This subsequently begs the question: how do we create and build teams that place our students in the best position to be able to accomplish the educational goals set by this type of education? This can be divided into two aspects. The first one relates to team compositions and how teachers need to be able to create balanced teams in terms of skills and abilities, as well as interpersonal and transversal skills. And a second aspect is team building, where trust is built among team members in order to create team cohesion and promote collaboration.

4.1.3. Paper I

Design team formation using self-assessment and observer-assessment techniques; mapping practices in a global network of universities.

As previously mentioned, a key element of the educational model's process is teamwork. Teamwork is essential for the projects and for achieving the desired learning objectives. To a large extent, I have presented in this thesis the importance of learning designerly ways of knowing and doing; however, the educational model not only provides

a platform for design learning but also provides a context for other multiple lessons that come from cross-disciplinary collaboration. Teams that have a well-balanced distribution of skills and capabilities that can also cooperate towards obtaining the common goal have a better opportunity to achieve both learning and project goals. On the other hand, poorly constructed teams may experience infighting, reduced performance, uneven contributions, or fragmented work without collaboration.

With the study conducted in Paper I, the intention was to examine students' assessment and team formation practices from different teaching teams within the network. The main objective was to be able to contrast different tools, methods, and procedures as well as theoretical underpinnings used to discuss and contrast current practices with literature. Based on this, we can discuss this paper from a descriptive perspective of the practices to inform and enhance understanding of team creation procedures rather than a prescriptive study of 'how to' create teams.

Contribution & Analysis

Toward the educational model

Based on the theoretical exploration of team creation that is connected to team creation in education, one of the relevant points that contribute to the development of the educational model pertains to the use of assessment methods to yield certain criteria for team creation. As explained by Estes et al. (2013) in discussing the different approaches to team creation, they highlight the importance of the use of 'intelligent' methods of team selection. They discuss in their finding that non-intelligent methods, such as randomized grouping or self-grouping methods, yielded less effective teams than those in which an 'intelligent' method was used to determine individual traits

and use them to group students.

One of the common methods that have been utilized not only in academic settings is the use of self-assessment personality tests, such as MTBI, in order to identify certain desirable characteristics of the individuals that can be paired with other individuals with different characteristics. However, the use of self-assessment has also been somewhat disputed, given its reliance on self-reporting and the possibility of misrepresentation of the actual characteristics. Furthermore, in the case of MTBI, questions of validity, scientific rigor, and oversimplification have been raised regarding the personality categories. On the other hand, there is also the use of observer-assessment methods in which an outside party, such as a teacher or staff member, utilizes different activities and techniques that are aimed at revealing the participant's skills and behaviors.

In regard to the contributions of the study towards the educational models and to other practitioners and teachers, it is important to note that even though an 'intelligent' method of selection is better than none, the use of mixed methods of criteria and skill assessment provide educators with the best possible analysis to inform team formation. Yet, there are tradeoffs when using these mixed methods. Self-assessments in which the students respond to personality quizzes and are sorted into categories require less time and resource investment, but the results of the assessment are less reliable. While observer-assessment provides the educator with the best indications of the individual's abilities and behaviors, it is time-consuming and requires the development of experiential knowledge to properly identify them.

Furthermore, the study provided interesting insights regarding the underlying behaviors and practices of educators who utilize observer-based methods. Among the findings, we can highlight the importance

of the development of connoisseurship and expertise that educators need to develop in order to properly identify the intended criteria and recognize which combinations of student skills and behaviors are more suited to each project. So, the selection of activities used by the educators plays a significant role, given that they allow the teachers to experiment and test the hypotheses of both the individual students and the potential groupings before major decisions are made.

Team formation activities and tools as Boundary Object

From the perspective of boundary objects, we can see team formation from two perspectives. Firstly, from the perspective of activities or processes that are used by the educators in which students and educators interact. As discussed in Paper I, the activities used by educators go beyond questioners that provide a simple answer regarding students' abilities. These activities are often practical exercises in which not only the students are being observed for assessment but also important elements of the future projects are being communicated, such as methodology, process, and expectations. Furthermore, it allows the students to communicate their skills and abilities to the educators in a non-verbal manner. Furthermore, if we look at the activities based on its intrinsic value of having:

- Abstraction: it allows educators to observe and translate actions from the participants into traits and behaviors that can be used to construct teams.
- Accommodation and Modularity: Given that the activities do not have a single purpose but also serve as project communication tools, they allow for other aspects, such as methodologies, to be shared and recalled later in the projects.
- Standardization: and it allows the activities to be easily interpreter by both students and educators as the process goes.

In the same way that the projects have been described as a multidisciplinary combination of students, Paper I also helps to highlight that at the helm of projects like these, there are typically teams of multidisciplinary educators that collaborate in teaching and managing the process. Decisions regarding what traits have been observed and how to combine them into student teams are commonplace among these groups of educators. These collaborative efforts allow for multiple perspectives to be brought into team formation and the reduction of biases when trying to assess students. Based on this, the second perspective in which we can analyze the assessment activities is from the perspective of a coordination and communication tool for groups of educators to come to agreements regarding observed skills and behaviors assigned to the individual students. And as a negotiation activity in which teams and combinations of teams are proposed and discussed in order to arrive to a desirable combination. This allows us to then look at the activities based on its intrinsic value of having:

- Abstraction: That allows educators with different perspectives to negotiate and have a dialogue regarding their observations and their hypothesis of possible combinations.
- Accommodation: It allows to be used for both a dialogue of the observed traits and corroboration or disputation of these observations, but it can also be used in the proposal of teams and possible groupings.
- Modularity: In which the different parts of the activities can be used individually to evaluate either single observations or generalized reflections of the observations
- Standardization: and finally, as it provides a common platform in which the same actions can be interpreted by the different educators with different perspectives.

4.1.4. Paper II

Into the woods; building trust in teams through scaffolded adventures.

First, it is important to specify that the study was conducted outside of the educational platform of the Design Thinking and Multidisciplinary Projects course. Rather, the subject of study was a different program, the Strategy and Management in International Organizations (SMIO) master program, in which I, as well as other members of the DaMD teaching team, are also involved. Moreover, some students from this master's program also take part in the DaMD projects as part of their master's education. The activity developed in the master's program represents an interesting case of study that could yield important learning points to be utilized and implemented in the team-building processes of the DaMD student teams.

The main objective of the paper was to examine the excursion day activity within the SMIO program at Linkoping University and contrast and analyze it with the theoretical frameworks of trust-building. The aim was to identify the mechanisms that could serve as pedagogical tools for fostering trust bonds that are crucial for effective teamwork. This by reflecting on both the individual activities as pieces, the part that they each played in trust building, and how the complete event as a scaffolded event as a whole lead to building said trust.

Contribution & Analysis

Toward the educational model

One of the important observations that led to the development of this paper was that of team dynamics that happened during the course of

the DaMD projects, as well as in other sibling courses as part of both the ME310 network and the SUGAR network. As presented in the previous studies, teaching teams place effort and importance on the creation of design teams by matching skills, abilities, and behavioral attitudes to create "good" teams. However, it is not uncommon to see that these well-crafted teams also experience difficult times because of disagreements and conflicts during the development projects. This observation, in combination with literature on team building and trust building, lays the first important contribution of this paper to the educational model, which is that putting together a group of qualified individuals that complement each other abilities and behaviors is not enough to set up successful teams, and bonds and trust need to still be developed.

In the paper, we examine a specific event in which multiple activities are undertaken in order to create swift trust, which serves as a platform of already established trust that can further develop through later interaction during the course of the program. Each individual activity utilizes a different mechanism to engage the participants in trust building that can work individually or as part of a set of activities such as the one examined here.

Another important contribution of this paper that can be linked to the development of team and trust-building activities in the frame of the educational models is the development of swift trust through the use of scaffolding and adventure-based games to help develop trust faster. In accordance with trust-building literature, swift trust is created in specific types of situations, and where individuals have no previous bonds, it can serve as a starting point in which further later interactions can deepen the bond. Thus, creating an early activity that utilizes elements of swift trust in the frame of ludic activities in

which the main object of trust building is hidden under a meaningful game/activity that engages the participants.

Team building activities and tools as Boundary Object

In the context of the team-building activity hosted by the SMIO master program, there are several activities within the overall activity, so not all can be observed and discussed as boundary objects. The SMIO program is an international master's program at Linköping University, in which students from multiple nationalities and with different educational backgrounds from different parts of the field of management sciences come together. Given the widespread backgrounds of the participants, the excursion day activity was created as a team-building exercise.

Among the activities, we can find two specific ones that can be analyzed from the boundary objects' point of view: the scavenger hunt and the raft building. The intention of both activities is to serve as an exercise where team members need to collaborate, coordinate, and come to shared agreements in order to successfully accomplish the task. The design of the tasks is such that they cannot be executed by a single member or divided into pieces and then put together; rather, the participants need to interact with each other and find a solution by coordinating.

A final important element of both activities lies in the 'consequences' that not collaborating can bring, in which the raft exercise literally means falling into the lake. According to this, and in the context of boundary objects, we can discuss both the activity and the tools used in the execution of the activity as artifacts and processes that enable coordination and negotiation between participants with no previous history of collaboration. The boundary objects serve as

ways of interacting with each other, founding common ground in collaboration and communication while building trust. Furthermore, if we look at the activities based on its intrinsic value of having:

- Abstraction: The clear-cut goal of finding the objects and crossing the lake serve as simple elements that can be understood from different perspectives but as common goal in which dialogue is facilitated.
- Standardization: As the same activity and goals can be easily interpreted and used by participants from different nationalities and backgrounds, and then those understanding are negotiated in terms of the strategy to follow to achieve the goal.

4.2. Divergence Phase

As previously discussed during the introductory chapter of this thesis, once the projects commence and teams have been formed and given the design brief in which the challenge is presented, the process that follows is one that is divided into seven different missions that guide the students through a scaffolded series of steps. This can be further divided into two distinct parts: a first one that focuses on problem understanding and a second one on solution creation. Both the part of the project where the main objective is to mainly focus on problem exploration (divergence phase) as well as the solution development part of the project (convergence phase) are aimed at developing the students' abilities of designerly ways of doing in a practical and experiential way.

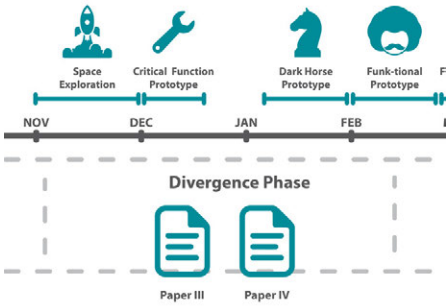


Figure 19. Divergence Phase

Both phases are underpinned by the designerly ability to explore the problem through the use of temporary solutions or prototypes followed by a reflection and a reframing of their significance to the understanding of the problem being solved and what a possible solution might entail. This is done in the frame of both reflection-in-action and the design thinking iterative process in which the students define the problem, generate, and ideate possible solutions. These possible solutions are constructed and tested, and the learnings are used to reflect on and to inform their next moves.

However, the difference between the divergence phase and the convergence phase lies in the focus on what is being reflected. In the divergence phase of the project process, the main focus is to reflect on the meaning that the early temporal solution has on the problem at hand and to enhance the team's understanding of their users, the needs, and the challenges that they face.

During the divergence phase, students go first through a purely problem-exploration mission and then through two prototyping missions. During the problem exploration mission, they decompose the problem, do field studies, and talk to users within their context. The first prototyping mission is aimed at establishing the critical elements of the problem, and the second mission is designed to explore divergent and 'dark horse' ideas that stretch what is possible within the problem area. In between the transition from divergence to convergence phase, students engage in a prototyping mission in which elements learned from the initial mission are brought into a rough first representation of a possible solution.

Within this process, there are two main activities that students do that are usually challenging. Firstly, we find the ideation process in which they need to produce divergent ideas from existing artifacts or solutions and in which they become easily fixated. Secondly, we find the creation of prototypes that do not aim at definitely solving the problem but rather as forms of testing hypotheses that can be later discarded. The main limitation of these two activities in the frame of a divergence phase is the difficulty that participants face when producing ideas and prototypes that are only meant as exploratory or proving actions and not as solutions proposals. Based on this, the following papers explore ideation and prototyping in the early stages, focusing on problem understanding.

4.2.1. Paper III

This is How We Roll, a play on creative thought and the generation of novelty.

In the early stages of the design process, when working with wicked or ill-defined problems, one of the important processes that designers undertake is to find ways to provide some structure to the problem through the proposal of temporary solutions (Archer, 1979; Cross, 1982; Darke, 1979). In design practice expertise (Dorst, 2015; Schön, 1983), part of the designer's ability is to observe, reflect, and re-frame the problem at hand, providing it with more structure but also allowing for new solutions to be proposed. However, even in the context of high levels of design expertise, being able to discard potential ideas or frames can be difficult, given our propensity to become fixated on past knowledge, experiences, or solutions (C. Cardoso & Badke-Schaub, 2011).

In the case of our educational model, we were able to identify that during the early ideation processes, where deeply divergent ideas to challenge and expand the understanding of the problem needed to be produced, students often struggled to get past their initial fixations. Based on the problem definition present in the brief and their own preconceived knowledge of existing solutions, students struggle to find new and divergent ideas for possible temporary solutions to challenge and expand said problem definition. With this observation in mind and in combination with design fixation literature, we explored the idea of developing a tool in the shape of a game that utilized randomness to force ‘interference’ and ‘context change’ in their ideation process to bring unexpected new domains to their existing problem exploration. By bringing these unexpected new domains, the teams were then asked to reflect on what these deviant

new ideas might mean for their problem and how it might help re-frame what a solution might mean.

Contribution & Analysis
Toward the educational model

From the perspective of the educational model, the first contribution to discuss here is the use of the tool to help students avoid design fixation and functional fixedness in their design process by forcing them to consider alternative uses or alternative contexts in which a solution might be used. In the context of early problem exploration, it was difficult for students to "let go" of their ideas of what a solution might be, which, in turn, restricted their ability to re-frame the problem. Furthermore, given that identifying the sources of fixation or even recognizing that one is fixated can be difficult, the game itself provided a useful platform to preemptively provide new scenarios and context to incorporate into the challenge.

The second contribution that can be discussed is the use of randomness as a way to aid the ideation process that fosters aimless and unguided exploration by forcing the students to connect elements that might be outside of the domain in which the problem is located with the elements that are. These are connections that otherwise might not be made by more inexperienced designers and that the use of the game facilitates. Furthermore, by providing these random scenarios for which the participant had no control, it allowed them to get past the initial self-consciousness and reluctance to share ideas or scenarios that might seemed not viable enough to be shared.

Ideation dice game as Boundary Object

When looking at the dice game as a boundary object that aids in the process of ideation within the context of the multidisciplinary

project environment, we can perhaps see a clearer connection between the game and the literature on boundary objects in the design process. The tool itself was created with the intention of being a tool that helped the groups of students to propose, communicate, and negotiate possible outcomes of the project and possible avenues to pursue under the disguise of the game. In this sense, the game can be seen as a multilayer process in which, with the help of randomness, students engage in ideation and the creation of deviant possible contexts. However, the important element of the game rested in the reflection that followed the ideation in which the scenarios were interpreted and abstracted, which then led to the presentation of new frames. The activity of reflection then served as a common platform where the team could discuss the implications of said ideas in the future of their project. Based on this, we can then see how the game can be seen as both an artifact and as a process in which coordination, transparency, and negotiation can be reached and boundaries of problem understanding can be crossed. Furthermore, if we look at the game based on its intrinsic value of having:

- Abstraction: the subjective nature of the story cubes allows each participant to interpret and create interesting new contexts while a dialog can be open by then identifying what could it mean for problem understanding.
- Accommodation and modularity: One of the key elements of the game is that it can be utilized during different session and different points of the project and in which interesting new combination can be had.
- Standardization: The simplicity of the cubes and the stories produce allows for the participants to immediately engage in creation of stories that provide contexts and then it promotes dialogue around those contexts into abstractions.

4.2.2. Paper IV

What is your (freaking) problem? Prototypes for problem exploration on the early stages of design

The use of prototypes in the practice of design is a key element of the designerly ways of both thinking and doing. Prototypes are the way in which designers explore both the problems that they are solving as well as the means to propose solutions that can be tested against the framing of the problem ((Cross, 1982; Darke, 1979; Dorst & Cross, 2001; Lawson, 2006)). This creates a feedback loop in which the temporary solution and its creation and testing creates new knowledge of the problem and helps to re-frame it (Cross, 1982; Dorst, 2015; Paton & Dorst, 2011). However, prototypes and our understanding of their meaning and how they are utilized differ greatly as their use is not exclusive to the design practice. While in engineering practice, prototypes are commonly associated with artifacts that showcase a high-fidelity representation of a solution in the later stages of the process, in design, their use is broader. From the design literature, as described in Ruiz & Wever (2023), prototypes are described as

"Embodiment[s], regardless of shape or medium, either physical or virtual, of a design idea with the intention of providing the designer a means to expand their understanding of the design space (BenMahmoud-Jouini & Midler, 2020; Camburn et al., 2017; Houde & Hill, 1997; Lande & Leifer, 2009; Ulrich et al., 2008)."

From this definition, we can then identify two key components of the designerly use of prototypes. Firstly, the embodiments of ideas and the use of prototypes as a way to flesh out thoughts and hypotheses of possible solutions (BenMahmoud-Jouini & Midler, 2020; Camburn et al., 2017; Pei et al., 2011). Secondly, as a means of exploration of the

design space, it highlights a fluidity of both what it means to explore and, consequently, what and how objects aid exploration. However, when exploring design prototype literature, it can be seen that prototypes are mainly discussed in the light of being embodiments of ideas and as means of exploration mainly in relation to the solution to be reached (figure 20). This then led to the question of what it means to prototype solely with the intention of exploring problems and not with the intention of creating feasible solutions.

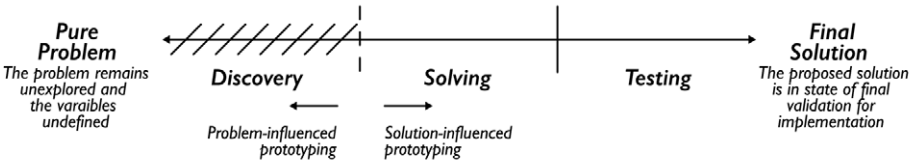


Figure 20. Prototyping for discovery

One of the main drivers for this study was the observation that in the early stages of design, some teams struggled to produce prototypes that were only meant as a way to explore the problem and not as a final solution. Another interesting observation that guided this study was that, however, other groups of students were not only able to create prototypes for problem exploration but said prototypes had a deep impact on the final solutions that they reached. With this in mind, this paper examined examples of student prototypes that unintentionally created prototypes that mainly focused on problem exploration and the used prototyping and reflection-in-action literature. This is with the intention of characterizing the point of reflection in which these prototypes placed emphasis on the differentiation between problem prototyping and solution prototyping.

Contribution & Analysis

Towards the educational model

The action of creating a prototype is very much associated with proposing solutions, and academic literature on the subject is a good indicator of this. The main intention of this paper was to provide a discussion point that centered on what the role of a prototype would be and that purposefully only focuses on the exploration of the design problem. In connection with the designerly reflection-in-action, we can discuss prototypes as moves or actions that can trigger a reflection based on the reactions caused by the move. However, the important aspect to examine is what is being reflected on. In a typical process of prototyping for solution creation, the problem context in which the solution is being tested has already been given structure or definition, and the requirements of a desirable solution have been established. Thus, the prototype and its verification is the action that is being evaluated, and the reflective action is taken upon the verification that said requirements of the design question are being either met or not met (figure 21).

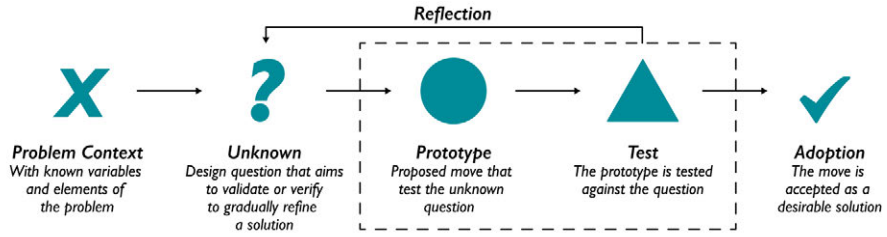


Figure 21. Solution-focused reflection-in-action

On the other hand, in the context of a problem exploration in the early stages, when the problem is still poorly defined or ill-structured, then we need to take a different look at what the move or action is. Based

on the concept of primary generators (Darke 1979) and the proposal of temporary problem frames, we can then take the temporary frame, the artifact created to evaluate said frame, and the evaluation of said artifact as the action or move. Consecutively, the reflection that is being made is against the problem context and what the temporary frame means for the structuring of the problem (figure 22).

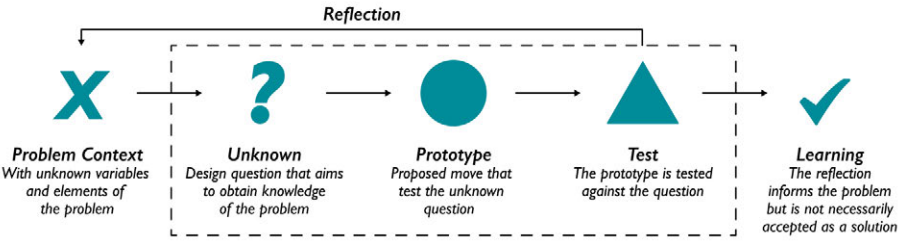


Figure 22. Problem-focused reflection-in-action

Based on the analysis presented above, the main implications that this study has in connection with the educational platform of design thinking and multidisciplinary projects are that of assignment planning and the role of the educator. An approach to prototyping for problem exploration needs a certain context and set-up in order for the reflection activity to happen.

Firstly, this kind of approach is deeply connected to problem structure and problem formulation. The problem context is that problem exploration prototyping needs to be one where the problem is ill-defined, where the educator has no knowledge of what the requirements for a solution are, and where the students are in charge of finding them. Secondly, the assignment and the deliverables that are provided to the students need to be created and structured around the reasoning of problem exploration. This means placing

emphasis on reflective action, learnings extracted from the active and problem frames as prototyping deliverables and not the artifacts that the students create. Finally, the educator plays a central role as well. In-class discussions, feedback sessions, and evaluations need to be aligned with the assignment and deliverables in order to stir the conversation toward reflection on the meaning of the prototypes and away from the fixation on the artifacts.

Prototypes as Boundary Object

Prototypes can act as tangible manifestations of ideas that bridge the communication gap between diverse stakeholders of the design process aiding to find common understanding (Carlile, 2002). Prototypes can help to demonstrate design concepts, requirements, and user experience expectations, which in turn facilitates communication and negotiation between groups or individuals (Bergman et al., 2007; Hussmo et al., 2022; Pierre Johnson et al., 2017; Rhinow et al., 2012). Furthermore, prototype fluidity of both interpretability and physical manifestation allows for engagement in meaningful discussions to explore and expand ideas, to both validate and iteratively refine problem understanding and what might be a possible solution. Based on this, we can then see that envisioning prototypes as boundary objects that facilitate coordination, communication, and understanding among stakeholders helps to fulfill the role of means of exploration that prototypes aim to have.

Similarly to the discussion of the contribution of the ideation dice game, prototypes in the early stages of the design process in the context of multidisciplinary education are more straightforwardly connected to boundary objects. Within boundary objects literature, in relation to engineering and design, prototypes are often named as prime examples of objects that facilitate the crossing of boundaries between

different groups. Prototypes are used as concrete representations of abstract ideas that help participants discuss, test, and collaborate to reach a common understanding. Furthermore, when talking about prototypes for problem exploration, their function as proving elements of the problem allows designers and users to test and evaluate different elements of desirability that, in the end, provide further insight into what might constitute interesting directions to pursue. Based on the above, prototypes can be classified as artifacts that, at the same time, can provide coordination, transparency, and negotiability in the boundary-crossing process and which, by their very nature, possess all the intrinsic values of boundary objects of abstraction, accommodation, modularity, and standardization.

4.3. Convergence Phase

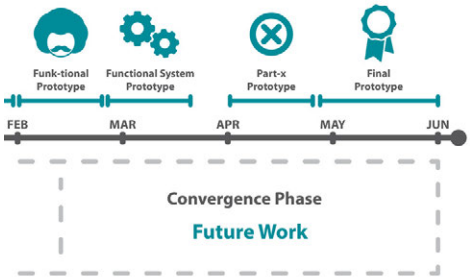


Figure 22. Problem-focused reflection-in-action

As previously explained in the section above the difference between the divergence phase and the convergence phase lies in the focus of what is being reflected on. In the convergence phase of the project process, the main focus is to reflect on the learning from the problem exploration and what it means to produce a solution to their given challenge. In practical terms, this process resembles more closely the usual process of a design engineering project or the process of

other educational models of design engineering like CDIO. This phase is characterized by the definition and narrowing down of the problem that is being solved and the definition of requirements and specifications for a solution that culminates in the production of a proof-of-concept prototype. During this process, the students go through 4 missions in which a final direction is set. From here on, each prototype advances both the concreteness of the solution and the resolution or fidelity of the prototype, going from rough and semi-functional to fully functional and visually refined. However, the work of my research and this thesis has focused on the early stages of the problem with a higher focus on problem definition and understanding, and other work that examines aspects of this phase could be of interest and perhaps discussed as possible directions for future work and future research.

05

Conclusions

Chapter 5

5. Conclusions

In the preceding chapters of this thesis, I embarked on a journey to present and discuss some of the problems and complexities of a problem-based educational model with a focus on the early design problem-solving process of problem exploration, understanding, and framing. As outlined in the introductory chapter of the thesis, the main purpose for this journey was to generate design knowledge that had a practical purpose and was readily actionable and usable to other educators in the context of multidisciplinary designerly education. With this purpose in mind, this section will then discuss the general contributions made by this thesis and how they add up to the fulfillment of this purpose. Subsequently, the answers to the research questions, the implications to practice, and the limitations of the studies are discussed in light of the findings and the literature. Finally, possible areas for future development and work are discussed.

5.1. Thesis contributions Answers to research questions

In the previous chapter, I examined the individual contributions of the different studies that comprise this thesis. In this contribution section, however, I will discuss the general contributions of this thesis as a whole. To accomplish this, I will utilize the thesis aims as a point

of reference to discuss the extent to which the aim has been fulfilled and how this aligns with the literature and to answer the research questions presented here.

Understanding problems

In the first aim of this thesis, a central aspect was to study practices and processes utilized by educators in problem-based designerly education to understand the underlying mechanisms and theoretical underpinnings of problem exploration in multidisciplinary education. Looking into the previous chapters of this thesis and what has already been discussed, one of the main problems that industry and professional engineering are currently facing is the growing 'wickedness' of the problems that need to be solved. This means problems are no longer simple and easily defined; rather, they are complex, poorly defined, and will initially present themselves to their solvers as interconnected messes (Ackoff, 1994). It is then the job of the problem solver to explore those messes and turn them into problems to which a satisficing solutions can be proposed. It is in this process of turning messes into problems and how we teach our students to deal with this process that we find the focus of my thesis .

In it, I have therefore explored the underlying mechanisms and theoretical underpinnings involved in understanding and framing problems in a problem-based educational context. This exploration has been framed from the perspective of educators' practices in problem-based learning, the methods, processes, and tools they use, and the problems they encounter in their practice. This framing, in my studies, has allowed for a designerly research process of taking the practical problems of the practices and studying the underlying theoretical 'reason why' of these problems in order to propose practical designerly solutions. The exploration of these theoretical

underlying mechanisms then serves as a point of contribution to educators to inform their own practices, may it be within problem-based learning contexts or outside of them, that help to better plan and utilize the practical methods, processes, and tools. In practice, this meant an exploration of aspects or problems that educators in problem-based learning contexts encounter in the process of setting up and teaching in a problem-exploration environment.

Crossing multidisciplinary boundaries

The second aim of this thesis was to explore and discuss the same processes, methods, tools, and activities within them in the context of multidisciplinary education that help to bridge boundaries between participants from different disciplines. As mentioned before, problems are no longer simple and easily defined; rather, they are complex, poorly defined, interconnected messes, and they do not exist in the vacuum of disciplines, so a multidisciplinary approach to solving them is also necessary (Ackoff, 1994). However, the reality of higher education is that of separate fields of study, each with its own compartmentalized body of knowledge, which creates a knowledge boundary between individuals from separate fields. In order to collaborate effectively across disciplines, individuals need to be able to communicate, negotiate, and collaborate across this boundary. Thus, the problem that educators need to address in multidisciplinary environments is to provide tools and methods that allow for the crossing of boundaries to facilitate the problem-solving process and, at the same time, the attainment of the educational goal of learning to collaborate and cross boundaries with individuals from other disciplines.

Based on the previous, the intention behind the inclusion of boundary objects in this thesis was to explore and discuss how the use of our

designerly objects can facilitate communication and collaboration between students, educators, and other participants and how those objects enhance our educational practices. In order to examine each of the individual objects, several dimensions and aspects of the boundary objects literature were utilized to frame the different characteristics that the object possessed. Several of the tools discussed here have already been presented based on certain characteristics that contribute to their use in the educational platform. However, intentionally or unintentionally, these objects often play more than a single role in the educational process, in which their role as a boundary object can be seen as a secondary yet equally important function. In using boundary object literature to the designerly objects used, we can once again highlight the different underlying mechanisms of each object in their multifaceted functions within the educational context.

5.1.1. Research Question 1

> ***RQ1:*** *How might we as educators provide our students with problem framings in the form of design briefs that provide them with a starting point framed in a way that fosters problem exploration? How can the brief and the process of creating it serve as bridges and boundary-crossing artifacts that aid in communication, negotiation, and collaboration between students, teachers, and other stakeholders of their educational process?*

Understanding problems

The problem that led to the first research question was based on observations of the problems that students encounter in their initial grasping of the problem and how to start the problem-solving process; I studied design briefs and problem formulations and

the implications that different problem framings can have in the problem-solving process. Design briefs serve multiple purposes in problem-based education. Briefs serve as the first point of contact that students have with the problem that they will be facing during their educational process. As points of reference and evaluation that can be expanded to mean new things based on problem exploration or as a source for decision-making in evaluating possible solution paths. And as communication tools that allow for collaboration and negotiation between multiple parties, such as student to student, teacher to student, and teacher/student to other stakeholders, such as commissioning organizations. Given the multiple uses that design briefs have in the educational problem-solving process, they represent an important element that needs to be carefully taken into consideration in order to allow for a problem-solving process that is conducive to the learning objectives. Based on this, I did an exploration of the different elements of design briefs based on past project briefs in order to identify different aspects of problem definition in accordance with the literature. Then, a typology of problem formulation was created and discussed based on the different elements of each type and literature on problem-solving and the potential implications that they have in the problem-solving process.

Crossing multidisciplinary boundaries

The use of the design brief and how the problem is formulated plays a more significant role than just being a static informative object to all the multiple stakeholders. The design brief as a boundary object, in the context of this thesis and of Problem-based learning processes, serves a rather crucial role in negotiation, communication, and coordination tools. Firstly, by allowing educators and organizations to reach agreements that satisfy the educator's educational goal and

the organization's desired outcome goal. Secondly, by providing a means of communication and clarification of the project, the existing knowledge, the limitations, the expected goals, etc., between the commissioning organization and the students. Finally, it is an object that helps students of the team to discuss interpretations, clarify meanings, and collaborate in the development of the projects.

Additionally, part of the licentiate thesis outcome was the creation of a tool for brief discussion, negotiation, and creation based on the theoretical aspects of problem definition and problem-solving processes. This tool was intended to aid in creating briefs based on knowledge of the initial 'mess' and the desired paths and outcomes of the process. The tool uses aspects of what is known about the problem space, what is known about the possible solution space, and what is the desired goal of the problem-solving process. This information is contrasted with what is known about the project setup. Based on this, educators need to first take into consideration the context of the course in which the students will be solving the given problem. Elements such as time allocation to problem and solution exploration, level of resolution of the final outcome, level of expertise of the students, and expected innovativeness of the solution play a part in deciding the scope and definition of the design brief. In the context of an innovative PBL or CBL educational model where the expected starting point is very open and ill-defined, there is still a need for educators to carefully craft problems that students can grapple with and where sufficient information is provided. The tool can also allow the negotiation of expectations, processes, and outcomes with organizations that sponsor projects in order to align their goal for commissioning or sponsoring such a project with the course objectives.

5.1.2. Research Question 2

> ***RQ2:*** *How might we use existing design methods, such as ideation and prototyping, to help students better define, understand, and reframe problems through temporary discardable solutions? How does using ideation and prototyping methods that focus on problem exploration serve as boundary object artifacts help facilitate these processes and bridge disciplinary boundaries between students, teachers, and other stakeholders?*

Understanding problems

The problem that led to the second research question relates to exploring poorly defined problems in order to provide them with a clearer definition and the role that design tools and methods have in the process of exploration. In the context of designerly problem-based learning, the early stage of solving a process involves exploring the problems by using temporary solutions that serve as tools to test and redefine what is known of the problem in order to reframe it. However, in the context of teaching designerly ways to non-design practitioners, it can be hard to steer away from the linear process of solving problems they have acquired in their specific discipline. In practical terms, this means that in the early stages, many students struggle to explore deviant or unusual framings and to discard temporary solutions, becoming fixated on previous notions of what the problem is and the early solutions they produced.

In order to examine this problem, I explored the underlying theoretical mechanism behind design fixation and the negative impact that it has in the production of new unforeseen framings. Furthermore, I explored the designerly process of reflection-in-

action as an underpinning concept in order to discuss how project tasks are framed and the points of reflection that the students are asked to address. In practical terms, this meant firstly an exploration of an ideation method with the introduction of randomness as a way to force the participants to create new ideas in which they had to include outside elements that were out of their control. Secondly, an exploration of how prototypes can be used as solely as a means of problem exploration. Based on both these exercises, I discuss the importance of the reflective process that follows the idea generation and prototype-building activities as the important aspect that allows them to be used as problem-exploration tools.

With the studies in Paper III and Paper IV, the intention was to examine the practices used and developed in our educational platform from the practical and theoretical perspectives. Based on this, we can observe from both studies (described in section 4.2) that the framing of the ideation and prototyping activities, the intended outcome/deliverable, and the reframing of reflection action can lead to the exploration of problems. Intentionality in the use of the method and how the deliverables are framed by educators are then key aspects in successfully utilizing the tools and methods. Moreover, the use of external aids, as described in Paper III and in section 4.2.1, elements outside of the control of the participants accompanied with instructions on the point of reflection, can serve to stir them towards problem reframing and away from merely the solutions. It is important to clarify, and perhaps propose a direction for future studies, that the ideation tool and the introduction of randomness have proven effective in aiding the participants in generating innovative and deviant ideas. However, the tool itself, without the conceptualization process, the reflection in light of the problem, and the facilitation of educators that direct the process towards problem framing, remains just a tool to create random ideas.

In terms of the prototyping for problem exploration, the same finding remains applicable. The process of problem prototyping, as explained in paper IV and section 4.2.2, does not differ significantly in its creation from a normal prototyping process. Ideas for possible temporary solutions to the problem are proposed, built, and tested. However, the initial instructions that focuses the effort, the process of reflection, the subject of the reflection, and the facilitation role of the educator do differ significantly and is what drives problem exploration.

Crossing multidisciplinary boundaries

The activities and objects used in both processes play a significant role as boundary-crossing objects by facilitating communication, collaboration, and negotiation of problem framings in the student teams. The use of the activities serves as a point in which the group intentionally communicates, negotiates, and creates framings with the help of the ideation cubes and the prototypes as reference points to guide the conversation.

In the case of the dice ideation technique, the use of the objects during the idea-creation process facilitates communication and collaboration between participants by providing a space where new possible frames and stories are created jointly in the same space. However, the most important function of the ideation technique as a boundary object occurs during the processing of the stories to turn them into new problem framings. In this instance, participants discuss and negotiate what each story might mean and how it can be abstracted as a different framing for possible solution space.

In the prototyping activities, it is perhaps easier to understand how the creation of a prototype serves as a collaborative activity in which the participants negotiate and build artifacts that provide a solution

to the problem. However, once again, as in the case of the ideation technique, the most important boundary-crossing function of the early prototyping happens in the processing of the results of the prototyping testing. During this processing activity participants reflect, discuss, and come to a shared agreement on the significance of the prototype testing and the implications it has moving forward. And so, the boundary-crossing object in this case is not the prototype itself but rather the mission/assignment protocol that forces the discussion and negotiation.

5.1.3. Research Question 3

> ***RQ3:*** *How might we create teams and build team dynamics that represent real-world diverse and multidisciplinary team situations and are conducive to achieving the learning objectives? How might we construct and use boundary object processes to help bridge team creation between educators and as processes to develop team dynamics?*

Understanding problems

The third observed problem that led to the third research question is related to what has been previously discussed in this thesis on the importance that a multidisciplinary approach has in addressing 'wicked' problems. A key educational goal of multidisciplinary educational models is to also prepare students to work and collaborate in teams of individuals with different backgrounds and perspectives, which is typical of the current working environment in organizations. However, teaching in multidisciplinary environments also comes with different problems that educators need to address and prepare for in the context of problem-based educational contexts.
The first problem that I explored was that of team creation, as putting

together cohesive teams of students from multiple disciplines that resemble a real working environment plays an important role in creating an environment that is conducive to reaching the learning objectives. Here, the challenge for educators lies in identifying students and student team combinations that have characteristics that are important for the problem-solving process, but that also complement each other in creating a cohesive unit. Furthermore, the second problem that I explored was that of not only creating cohesive teams but also building bonds within the student team in order to foster better collaboration in the process. To do this, I first explored the theoretical aspects of team creation and the role that discipline-specific skills and interpersonal transversal characteristics of individuals have on team performance. Second, I explored the different mechanisms of characteristics assessment and the trade-offs that different assessment processes have in correctly identifying relevant aspects of behaviors and skills.

In this exploration, tensions and trade-offs between the two approaches in the process of team creation in the educational context were discussed. Educators often have to negotiate the trade-off of time investment vs. more data to inform their decision-making. While self-assessment methods are less time-consuming, the validity of results can be questioned, given the possibility of misrepresentation in the answers provided by the participants. While observer-assessment is more time and resource-consuming, educators can verify the desired characteristics that are being looked for. Furthermore, observer-assessment methods rely more heavily on the observer's expertise and ability to observe and verify said characteristics. Based on these observations, a partial answer to the question is that educators, based on the context in which the characteristics are being observed, must make decisions and accept the trade-offs of choosing assessment methods. The use of mixed methodologies can aid in managing

tensions and trade-offs, where self-assessment helps in the mapping of characteristics, and observer-assessment helps corroborate these.

Finally, I explored the theoretical aspects of team bonding and trust in teamwork, the mechanism through which trust is built, and the process that can help to 'accelerate' the building of trust in special circumstances. Moreover, we can also draw from the literature on the importance of building trust within teams that aid in the construction of team bonds (Hero & Lindfors, 2019; Jonassen et al., 2006; Kurtzberg, 2005; Van der Vegt & Janssen, 2003). In order to provide an answer to the question, we examined the use of an intentional scaffolded exercise that leverages an adventure game-like approach to generate swift trust among students. The key element of the findings rests on the use of a process that builds early, swift trust (Meyerson et al., 1996). that can serve as a starting point in which trust can be built later on in the project process.

Crossing multidisciplinary boundaries

In both processes of team formation and trust building, the artifacts and processes used play a significant role in crossing boundaries between educators and students. Firstly, in the process of team creation, both in self and observer assessment, the objects used play a central role as points of discussion and mediation in the team creation decision-making process. Given the multidisciplinarity of the projects, it is not unusual to encounter that teaching teams that run this kind of educational model are also comprised of a set of multidisciplinary individuals with varying levels of experience. These teams of educators, with varying observations and points of view, use these objects as points in which perspectives and observations were communicated and negotiated in order to arrive at a consensus. In which case, the process of negotiation and coming to an agreement on which student and

which team combinations are more desirable are as important as the object in which the different criteria for team formation are observed.

Secondly, in trust building, multiple boundary objects facilitate communication and negotiation between individuals and become the conduit through which trust bonds are formed. As discussed in Paper II, trust develops through multiple points of interaction between individuals in which they place confidence in one another based on the mutual willingness to cooperate and collaborate with others for a common purpose. In the case of the excursion event and the different activities within it, the intention is to engage students in cooperative activities in which, in order to be safe inside caves, not get lost in the woods, and cross the lake, trust, communication, and collaboration are indispensable. Thus, the objects that simulate stressful situations where swift trust is built serve as communication, negotiation, and common action enacted, which further help the trust bond to be created.

5.2. Implications

The outcomes and contributions of this thesis carry implications for our theoretical understanding of the tools, methods, and processes used in the educational platform and its practical applications. Moreover, it can also be argued that the contributions and findings of this thesis have the potential to shape the strategies and decisions made by educators in other similar existing courses/programs or in the development of new similar ones.

5.2.1. Implications for our educational platform

The outcomes from the different studies and research presented in this thesis have already provided immediate practical implications in the curriculum of the projects and courses associated with the

educational platform. One of the important elements that have been central to the teaching team of the DaMD projects at Linköping University has been the use of the same principles of experimentation and iteration that the course teaches. And so, based on the findings and outcomes of the studies conducted, several of the approaches and methods that we ourselves utilize have already been shaped based on those findings. Furthermore, given that the studies in this thesis have all sought to understand not only the methods but also the theoretical reasonings and underpinnings behind the methods, several of the methods themselves have been iterated based on the foundational underpinnings. Problem formulation practices and discussion, team formation based on student assessments, randomized ideation sessions using the cubes, and problem-based prototyping remain part of the course.

In past sections of this thesis, it has been explained that the studies, with the exception of paper 6, were conducted within the practices of the educational platform as the center of my doctoral studies. In this sense, it is very natural that the outcomes of the research have practical and theoretical implications for the teaching of the projects. However, the same outcomes have also provided learnings that have percolated to other educational contexts and courses in which I also participate. The outcomes of the licentiate thesis have provided a background to problem framing that has been used in a strategy course for opportunity detection and market creation. Furthermore, the ideation story cube tool has been used in several other courses with the same intention as in the DaMD course. It has also been used as an assessment tool to observe students' abilities to abstract from the stories to possible frames and their willingness to playfully engage in the activity. These two examples can also serve as indications that the tools are not necessarily bound to the educational platform but can also be utilized in other educational contexts, given the studies focus on the theoretical underlying principles.

5.2.2. Implication for others

Beyond the confines of our specific educational platform and the DaMD course, the implications extend to the broader design engineering education landscape. Educators across institutions within the ME310 course and the SUGAR network, as well as other similar educational contexts that utilize CBL or PBL, can leverage our findings to reassess and enhance their own teaching practices, adapting their approaches to align with the identified learning of the studies.

Based on the thesis's aim of generating practical and actionable knowledge, the studies focused on examining both the specific practices and their underlying mechanism. With this in mind, an educator from other contexts can utilize the findings to, for example, enhance their existing prototyping assignments and deliverables to further place emphasis on reflecting on the findings to better inform problem exploration and re-framing. Aspects of their own student assessment approach and characteristic detection can be examined under the theoretical and practical findings of team creation. Additionally, the discussion and contribution of this thesis regarding the use of the practices and objects as boundary-crossing objects also provide insights and reflections into their own practices. Aspects such as how their own problem exploration boundary objects help to bridge knowledge gaps and how they help or hinder collaboration between the different stakeholders.

Finally, the findings of this thesis also have implications for design education in general as well that point to the importance of further development of students' designer skills of problem exploration and reframing. Educational processes that focus on problem exploration and framing in engineering education can be better integrated into

engineering curricula, and how they interplay with more solution-oriented processes remains an important area of discussion.

5.3. Limitations to implementation

While, as presented before, it is important to discuss and present the contributions and the implications of the papers presented in this thesis, it is important to acknowledge and examine the limitations that are inevitably a part of research and academia. The intent of this section is then to address the limitations of the studies presented here and the educational model as a designerly approach to engineering education, which is the core of the thesis.

5.3.1. Limitations of context, methodology, and generalization

The first limitation to present in this section relates to the generalizability of the findings. The findings of the studies here present are based on the specific context of the DaMD projects, the ME310 course, and the SUGAR network, which are based on the educational model and missions created by the ME310 course over an extended period of time. This means that the practices presented here have been examined in the context of these courses, which have already developed a methodological approach, core teams of experienced teachers, and an administrative support system in which the course operates. Furthermore, the studies themselves were mainly conducted within the frames of the already mentioned DaMD, ME310, and SUGAR contexts as PBL/CBL contexts. All of these do not necessarily fully represent all PBL/CBL environments, but given their history and record of academic participants, methodological development, participating organizations, and project output, they do represent a good case of study f a kind of PBL/CBL context.

Furthermore, the main intention of the studies discussed here was not to present a prescriptive answer or "how to" guide to any of the subjects of study. Rather, the intention was to examine the underlying mechanisms and effects of the methods and processes in order to provide usable and actionable knowledge for other educators and researchers. However, the translation of those underlying mechanisms and theoretical underpinnings back into actions in other educational settings might require the development or adequation of the context in which they can be applied. And so, educators should exercise caution when applying the results to different settings and contexts.

Furthermore, it is also important to mention that the use of both action research and research through design, which place a strong emphasis on participant experiences and context-specific findings, adds complexity and limitations in generalization. The use of both methodologies can produce outcomes that are often deeply tied to the specific environment, participants, and conditions of the study. With this very much in mind and as mentioned in previous sections, the intention of the papers and of this thesis was to examine underlying mechanisms and theoretical underpinnings to allow for more generalizable knowledge. Yet this context dependency remains a limitation that needs to be considered.

Moreover, from a previous chapter of the methodological approach, and based on Gaver's (2012) and Zimmerman et al. (2010) clarification, the design strives for the finding of a satisficing solution for the current context rather than a discovery of an absolute optimal solution or undeniable truths. And so, a further discussion of scientific validity and falsifiability under the traditional scientific tradition can become relevant. However, it is through the principles of presenting the conceptualized findings from the design answers into research answers in light of the theoretical underpinnings of the practice

and the conscious refraining from prescriptive definite answers that the limitations of the use of these research methodologies can be mitigated.

5.3.2. Limitations of the educational model

Beyond the specific studies presented here, and regarding the use and possible application of the educational model in other contexts, there are certainly advantages of the application of said model, which have already been discussed in previous sections. However, there are also limitations that can act as impediments to the adoption of such an educational model and the tools, methods, and processes presented in this thesis. A model such as the one presented here, as well as others that utilize innovative PBL or CBL, requires a shift in the educational approach paradigm that goes beyond the simple creation of a course.

Among the main challenges of the educational model is that of its implementation. Based on the topics discussed in this thesis, we can observe the limitations and difficulties of the model in regard to resource intensiveness, uncertainty and ambiguity, and educator expertise. Other limitations and difficulties of the model can also include scalability and class size, assessment and grading, and student adaptation based on prior experiences; however, this will not be covered since they were not part of the studies conducted for this thesis.

Resource intensiveness: One of the main limitations already discussed in several of the studies here presented relate to high demand that such model has on the teaching staffs time and commitment. Aspect such as projects acquisition, project planning and design brief research and formulation, assessment, and evaluation of students for team creation and so on, demand a high level of time commitment from the teaching staff.

Uncertainty and ambiguity: One of the core strengths of such education and the use of said tools lies in teaching students to deal with high levels of uncertainty and ambiguity in their projects and the possible results. However, it is also a limitation to teaching staff who, at the same time, have to deal with high levels of uncertainty and ambiguity in dealing with course development and project development. This can be better seen in the tension or careful balance between prioritizing the attainment of the educational goals of the students vs the expected project goals of the organizations. Where undesirable outcomes (failures) are seen as great learning points for students but perhaps not as much for the investing organization.

Educator expertise: The importance of the educator's expertise when dealing with many of the processes and methods discussed in this thesis has previously been discussed. In the process of planning the projects, expertise in developing a sense for the different types of problem formulations that students need in order to cope with project deliverables. In team creation assessment and observation of important criteria that might be relevant in team dynamics. Furthermore, in dealing with these types of projects, the role of the educators significantly shifts from knowledge provider to facilitator. In which case the role of feedback giving in areas such as problem framing, ideation, and prototyping learnings requires a shift in role. From knowing the answers and guiding students to attain them to not knowing the answers and guiding students to discover them themselves. Carefully balancing between providing assistance to the students while avoiding imposing their own ideas of what the answers ought to be.

5.3. Future Work

While the studies presented in this thesis have provided insights and understandings that inform some of the processes, methods, and tools used in our designerly educational platform, there is potential to extend the scope of said studies. Future research that expands on each of the individual studies could be of great interest. Further exploration of design brief, and problem reframing beyond the first and initial representation. Problem exploration prototypes in connection to the different types of brief and how each type fosters and facilitates this type of prototyping effort. The relation of team formation and building to problem exploration and how the team compositions, in terms of diversity, skills, and knowledge, affect how problems are explored could be areas of possible interest.

However, beyond the expansion of the existing studies, there is the possibility of exploring areas of the educational platform that have not been explored. In this thesis and the studies within it, one of the main objects of observation has been geared toward problem exploration and problem framing, which is situated in the early stages of the design process of the projects. This means that much emphasis has been placed on understanding and reflecting on the planning phase and the diverging phase of the educational platform. Consequently, aspects of the convergence phase that are more deeply connected to the creation and development of solutions have been left aside for future work and future possible areas of exploration. With this in mind, there are several aspects of the convergence phase that are of interest to examine. Here, I will present some possible areas of future interest not based on any order of priority or urgency but rather in the same project chronology order that I have used before.

One of the areas of interest for future exploration relates to the point of transition between the problem exploration divergence phase and the solution creation convergence phase. The scaffolded methodological approach that the projects utilize already uses mission and deliverables to facilitate the process, yet the task remains a difficult inflection point for the student teams. Furthermore, aspect of multidisciplinarity and international collaboration make said transition an even more cumbersome undertaking. Future work could examine the process from a practical perspective of successful and unsuccessful transitions. From a methodological perspective of the assignment and tools used to facilitate it. And from the theoretical perspective of the underlying behaviors and mechanisms at play that facilitate and/or hinder the process.

Another interesting area of future exploration relates to the development and framing of design requirements that mark the point in which teams select the 'final' direction in which efforts will be placed towards their final project deliverable. Within the educational platform missions chronology, the process requirement setting does not begin when the convergence phase begins. Students go through two more solution exploratory missions aimed at mapping more than one possible direction for solutions with low-fidelity and low-investment prototypes before requirements toward a final solution are considered. In the practices that we utilize in the DaMD course, aspects and methods of systems engineering are brought to the deliverable in order to facilitate the process of reflecting on all the different possible directions in order to facilitate integration. In the same manner that this thesis has examined the methods used in our course with the intention of highlighting the theoretical reasoning at play, this integration of systems engineering methods could serve as a future area of research that looks into the integration of such methods into a designerly education process.

Finally, another interesting area of possible exploration relates to post-project studies that could be related not only to project effectiveness and the evaluation of the results that are obtained from the project but also to the integration and implementation of the findings into the organization. One of the interesting aspects that were briefly mentioned in the licentiate thesis was that of organizational motivation in regard to participating in the design projects. Empirically and based on years of interactions with different organizations, it is implicitly known that organizations come to the projects with different motivations. The projects and the context in which they happen provide a wide range of possible outcomes that are desirable for these organizations. Outcomes such as the final results, the multiple unfinished ideas and avenues left unexplored, the multiple frames placed on the initial problem, the methodology, and its possible implementation in the organizations are some of the more commonly discussed motivators. Based on this, several interesting aspects of how all these outcomes can be evaluated, integrated, or implemented within the organizations could be of future interest in order to examine and discuss project effectiveness and contributions to the sponsoring organizations.

List of References

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Chapter 6

6. List of References

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Appended Papers

Papers

The papers associated with this thesis have been removed for copyright reasons. For more details about these see:

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"There is no real ending.
It's just the place where you stop the story."
Frank Herbert

"Vivir es como escribir,
cualquiera lo hace pero no todos lo dignifican"
Juan Carlos Fonnegra

Espero que con el esfuerzo de todos estos años
escribiendo esta tesis, haya dignificado a mi familia, a todos
aquellos que me han apoyado, a mi profesíon y a mi país.

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